

**Modular Open Systems Approach (MOSA) and its Impact on the US Defense
Acquisition Methods and Programs**

by Parth Devang Shah

B.S. in Mechanical Engineering Technology, May 2014, Rochester Institute of
Technology
M.S. in Manufacturing Leadership, May 2016, Rochester Institute of Technology

A Praxis submitted to

The Faculty of
The School of Engineering and Applied Science
of The George Washington University
in partial fulfillment of the requirements
for the degree of Doctor of Engineering

January 8, 2021

Praxis directed by

Tim Eveleigh
Professorial Lecturer of Engineering Management and Systems Engineering

Thomas Holzer
Professorial Lecturer of Engineering Management and Systems Engineering

The School of Engineering and Applied Science of The George Washington University certifies that Parth Devang Shah has passed the Final Examination for the degree of Doctor of Engineering as of November 13, 2020. This is the final and approved form of the Praxis.

Modular Open Systems Approach (MOSA) and its Impact on the US Defense Acquisition Methods and Programs

Parth Devang Shah

Praxis Research Committee:

Tim Eveleigh, Professorial Lecturer of Engineering Management and Systems Engineering, Praxis Co-Director

Thomas Holzer, Professorial Lecturer of Engineering Management and Systems Engineering, Praxis Co-Director

Amir Etemadi, Associate Professor of Engineering and Applied Science, Committee Member

© Copyright 2020 by Parth Devang Shah.
All rights reserved

Dedication

This is dedicated to my late grandparents, for without their hard work, resilience, and strength I would not be here today. Late Savailal Keshavlal Shah, Late Virbalaben Savailal Shah, Late Prof. Hiralal Bhavsar, and Late Sushila Bhavsar, I owe them everything for without their support, I would not be here today. This is for you, your legacy, the past, present, and future cohorts of the Shah family line of graduates.

Acknowledgments

I wish to acknowledge my advisors Dr. Tim Eveleigh and Dr. Thomas Holzer for their guidance, assistance, insight, and working tirelessly with me throughout my journey ensuring my success as I cross the finish line. I would also like to acknowledge some very important people in my life for all their moral support, understanding, encouragement, and sacrifices when I needed it most during my challenging times.

I would like to thank my father (Devang Savailal Shah), my mother (Sonal Devang Shah) and my brothers (Harsh and Max Shah) for being by my side and having the perseverance to always push me to do my best. I would also like to thank my fiancé Aashka Bhatt for challenging me to do better, dig deeper, finish well, and always be confident in my work.

I would also like to thank my extended family members – Jayesh & Nila Shah, Tejas & Ami Shah, Pinkesh & Pooja Shah, Priti Doshi, Priya Doshi Saha and my fiancé's family (Vina, Apoorva, Nimisha, and Pankti Bhatt) for their help and encouragement. I can never fully appreciate the kind gestures extended towards me, but I can emphasize how much it meant to me and my family. And last but not the least a few of my very dear friends for always having my back and supporting my dreams and aspirations – Nikesh Hajari, Jesse Radzikowski, Douglas Agyemang, Aashik Shamanur, Veerendra Raj, and Ronak Shah for keeping faith in me and supporting me to complete such a formidable endeavor.

Abstract of Praxis

Modular Open Systems Approach (MOSA) and its Impact on the US Defense Acquisition Methods and Programs

Almost 60% of the US Major Defense Acquisition Programs are either over budget or delayed in terms of schedule. To combat and address this, the Department of Defense has introduced and mandated the implementation of the Modular Open Systems Approach through the Better Buying Power 3.0 directive in 2015. This strategy will hope to combat technology obsolescence, upgrade current highly cohesive systems, and increase flexibility and innovation all the while trying to battle cost and schedule challenges.

The US Defense industry is arguably undergoing its greatest transformation ever. Under strong pressure to decrease costs, accelerate the speed of product development, promote innovation, and enhance quality, the US Defense Department has been searching for new methods to transform its defense acquisition process. The US Department of Defense is the largest defense organization in the world, responsible for spending about \$989 billion annually on technology acquisition. In fact, “the United States spends more on defense than the next nine countries combined” (Kimberly, 2019).

Several years in, there is no evidence showcasing that this strategy is successful. This research alone gathers this evidence, albeit largely qualitative, and suggests how modular and open system supporters and contributors can make its effects more effective so that this promising approach can be further sustained. This research will benefit students and practitioners of systems engineering, stakeholders across government, industry, and academia that are affiliated with the production, quality, and manufacturing fields to promote a collaborative vision of better acquisition processes and program success.

Table of Contents

Dedication	iv
Acknowledgments.....	v
Abstract of Praxis.....	vi
List of Figures	x
List of Tables	xi
List of Acronyms	xii
Chapter 1 - Introduction	1
1.1 Background	1
1.2 Research Motivation	4
1.3 Problem Statement	5
1.4 Thesis Statement	6
1.5 Research Objectives	6
1.6 Scope of Research	7
1.7 Research Limitations.....	7
1.8 Organization of Praxis.....	8
Chapter 2 – Literature Review	10
2.1 Introduction	10
2.2 Modularity	11
2.3 Reviews of the Major Defense Acquisition Programs from 2008 - 2019	12

2.4	Modular Open Systems Approach (MOSA)	14
2.5	Deeper Dive into Modular Open System Architecture/Approaches	17
2.6	Advantages & Roadblocks with Modularity and Open Approaches	22
2.7	Adoption of ISO/IEC/IEEE 15288 by the DoD to increase interoperability	29
2.8	An analysis of Modular Systems Approach in Avionic Defense Systems	31
2.9	Global effects of MOSA	31
2.10	Literature Review Summary	32
Chapter 3 – Methodology		34
3.1	Introduction	34
3.2	Hypothesis	34
3.3	Hypothesis Overview – H1 - Cost and Schedule Effects	34
3.4	Hypothesis Overview – H2 – MOSA as a successful strategy	37
3.5	Hypothesis Overview – H3 & H4 – US and non-US Government department impacts	38
3.6	Hypothesis Overview – H5 & H6 – MOSA and its impact on the commercial sector	40
3.7	Summary and Conclusion	41
Chapter 4 – Results		43
4.1	Introduction	43
4.2	Hypothesis Overview – H1 – Cost and Schedule Effects	43

4.3	Hypothesis Overview – H2 – MOSA as a successful strategy	46
4.4	Hypothesis Overview – H3 & H4 – US and non-US Government department impacts	49
4.4.1	US Army	50
4.4.2	US Navy.....	53
4.4.3	Avionics within the DoD	55
4.4.4	Global Overview	67
4.5	Hypothesis Overview – H5 & H6 – MOSA’s impact on the commercial sector	72
4.6	Summary of Results	74
Chapter 5 – Conclusions and Recommendations.....		80
5.1	Introduction	80
5.2	MOSA Discussion.....	80
5.3	Other Observations.....	83
5.4	Limitations	84
5.5	Contributions to the Body of Knowledge	85
5.6	Recommendations for Future Research	86
5.7	Conclusion.....	88
References.....		90
Appendix A – Cost Analysis of Major Defense Acquisition Programs of 2018		105

List of Figures

Figure 1: Major Defense Acquisition Programs 2008 – 2019	13
Figure 2: Pie chart for Question 1	19
Figure 3: Pie chart for Question 2	20
Figure 4: Pie chart for Question 3	21
Figure 5: Pie chart for Question 4	22
Figure 6: Successful Examples of MOSA within the US Army	51
Figure 7: Successful Examples of MOSA within the Navy	54
Figure 8: Overview of the Avionic Strategies	56
Figure 9: UCS Architecture	57
Figure 10: OMS Architecture Reference	61
Figure 11: Synopsis of the FACE Approach	65
Figure 12: FACE Integration Approach	66

List of Tables

Table 1: Survey Answers – Question 1.....	18
Table 2: Survey Answers - Question 2	19
Table 3: Survey Answer – Question 3.....	20
Table 4: Survey Answer – Question 4.....	21
Table 5: Summary of Hypotheses.....	76

List of Acronyms

ACAT	Acquisition Category
ASB	Avionics Service Bus
CAF	Canadian Armed Forces
CANES	Consolidated Afloat Network Enterprise Service
COE	Common Operating Environment
COTS	Commercial Off the Shelf
DoD	Department of Defense
FACE	Future Airborne Capability Environment
GAO	Government Accountability Office
GVA	Generic Vehicle Architecture
IEC	International Electrotechnical Commission
IEEE	Institute of Electric and Electronic Engineers
INCOSE	International Council on Systems Engineering
ISO	International Organization for Standardization
MBSE	Model Based Systems Engineering
MDAP	Major Defense Acquisition Program

MOD	Ministry of Defence
MOSA	Modular Open Systems Approach
NGS	Non-Government Standards
NPD	New Product Development
OMS	Open Mission Systems
OSA	Open Systems Architecture
SLOC	Source Line of Code
SOSA	System of Systems Approach
UAS	Unmanned Aircraft System
UAV	Unmanned Aerial Vehicles
UCS	Unmanned Systems Control Segment

Chapter 1 - Introduction

1.1 Background

The speed of change coupled with balancing emerging threats, technology obsolescence, rapidly evolving technology, and increased security requirements has required the U.S. Department of Defense (DoD) to change the way it acquires defense systems. The DoD has had to move away from traditional acquisition methods to embrace newer modular and open system methods. Recognizing the need for affordable solutions and achieving time and schedule targets, the DoD calls for the application of the Modular Open Systems Approach (MOSA). MOSA's purpose of preventing vendor lock, supporting affordable capability evolution, and encouraging innovation has made it an attractive method to acquire current and future defense systems. Vendor lock refers to restrictions that prevent the DoD from switching from one service provider to another. The company that built the defense systems holds the intellectual property rights for the system, thus can be the only service provider preventing switching to an alternate vendor.

In the US DoD, defense systems tend to frequently operate well beyond their initial planned service life and services use these systems in functions outside of their original scope or design. Systems that demonstrate some nonfunctional attributes such as performance, maintainability, and usability achieve platform transformability and continue to serve the DoD in spite of the changes to their operating environment. Nevertheless, the DoD does retire systems that cannot be upgraded due to vendor lock design constraints or affordability concerns or are unable to meet new functional requirements in new operational environments. Over time, as program and mission requirements change, threats increase and expectations grow, defense systems will be

expected to adapt be able to upgrade accordingly or systems that do comply to this will eventually be retired out of service. This is where modularity and MOSA play an integral part.

Today, many organizations are facing increased business uncertainty as they encounter innovative competitors. “Identifying new opportunities and strategically transforming to take advantage of these opportunities is a key to the long-term survival of a firm” (Ramesh & Delen, 2019). The United States is the top spender on defense and military equipment in the world and used to be known as the forefront in defense innovation. The DoD is responsible for spending over \$989 billion annually on defense technology through the employment of thousands of acquisition projects and people and thus not only survival but transformation is vital for it to grow. “The military enterprise substantively and directly affects the national economy and thus it is fitting to expect that military process improvement can have direct impacts on general civilian enterprises” (Badiru & Thomas, 2013). Currently, the DoD manages multiple projects. The largest projects managed are called Major Defense Acquisition Programs (MDAP) and projects of this category each meet or exceed \$480 million of total expenditures within a fiscal year. The United States Government Accountability Office (GAO) investigates and releases an annual report with respect to the cost and schedule performance of these programs and the impacts of cost and schedule overruns on these MDAPs. The GAO found an interesting trend when looking at the 86 MDAP programs (Appendix A). From these programs, the GAO found that the 61, that were all started before 2010 faced cost growth of 62.4 percent or \$542.1 Billion and schedule overruns of 35 months on an average. The remaining 25 programs that were initiated after 2010 observed a cost

growth of 2.1 percent or \$5.3 billion and schedule overruns of nine months as compared to the original estimates. Vital to note is that this trend may change over time as many of the programs after 2010 are now in the very early production stages or yet to enter the production phase. For a matter of perspective, in their 2017 report dated April 25, 2018, the GAO listed 86 major programs accounting for a cost a total cost of \$1.66 trillion. (GAO, 2018).

The research in the Praxis addresses the question of whether there is a relationship between MOSA and program success by critically examining the impact of MOSA on the defense acquisition process through various means. The Praxis begins with a focus on a deeper understanding of modularity and open systems, and their role as a strategy within the DoD. The Praxis will then further highlight case studies from commercial, non-governmental, and other global governmental department of defense lenses. There will be a deeper dive into avionic systems that forms a compelling example within the DoD where MOSA is actively being pursued in the advancement and development of modular systems and has experienced a high rate of success in integrating modularity. Consequently, this Praxis will focus on the standards of implementing modularity that are being developed and used in avionic systems. Lastly, this research will address the emergence of MOSA worldwide and its success leading to sponsored studies and partnerships by the UK Ministry of Defence (MOD), Australian DoD, and the Canadian Armed Forces (CAF). All of these organizations now advocate for the benefits of more open technical and business practices.

1.2 Research Motivation

The DoD needs to move away from the archaic system of buying hardware defense systems that do not communicate with one another and are specific to certain mission needs. While that strategy worked for decades, the game has changed.

When the Android platform was introduced, there was a lot of skepticism of its open-source approach. Fast forward to a few years later, the open system approach led to creative alliances of individuals and organizations to build applications that have become vital and are found in billions of personal devices. From maps to entertainment tools, there is an application for everything. The lessons learned can be applied to defense systems too.

To compete and be a leader today, there is a need for adoption of buying hardware-based defense systems to software-based systems that support real time upgrades through software. DoD's future success will lie on adaptable, open, and modular defense systems that are multidomain and can interact with multiple defense systems to provide an edge in decision making.

The US DoD has adopted the use of open and modular systems under a strategy called MOSA. Currently there is no specific study evaluating the cost effectiveness and other impacts of modularity or open architecture within the DoD acquisition strategies. There is also a lack of material showing strong evidence of modularity being a direct causal variable in the success or failure of a defense acquisition program.

Defense weapon acquisitions usually generate highly complex systems with multiple stakeholders. These stakeholders typically span from multiple locations and organizations collaborating to manage changes in requirements and scope. These efforts lead to

delivering major defense systems such as weapons, aircraft, submarines, satellites, outer space technology, and so on. The GAO is responsible for investigating the performance of these programs in terms of schedule and cost and their impact on the US Government. It is often observed that as systems get complex, there is a higher susceptibility for these programs to exceed their initial cost and schedule targets. “US Government Accountability Office (GAO) reports identified a range of possible root causes for the overruns from the structure of internal open approach policies for each of the military services (Army, Navy, Air Force), to the lack of expertise in facilitating an open systems approach” (GAO, 2013). The GAO’s 2017 report further helped identify whether adopting MOSA has aided in reducing these overruns or has led to a major impact. “Major defense acquisition programs (MDAPs) are notoriously prone to substantial cost overruns, and engineering and design issues are often partially to blame.”(Valerdi, Dabkowski, & Dixit 2015).

Obsolescence (of systems) is a huge problem costing the DoD an average of \$750 million annually - an alarming number. (Adetunji, Bischoff, & Willy 2018). Due in part to the lack of the knowledge needed to isolate the effects of modularity on programs and an inability to acquire detailed performance on MDAPs, there has not been significant analysis or studies performed with respect to modularity and its effects within the DoD. Studying the defense acquisition programs should allow a better understanding of the performance of defense acquisition programs over the years and help form a framework to better forecast costs and schedules.

1.3 Problem Statement

Despite a huge, multiyear investment in MOSA, its supporters are unable

demonstrate its value in a convincing way potentially threatening its continued use as a promising design and acquisition strategy.

1.4 Thesis Statement

An organized search for the representation of evidence of the success of the MOSA strategy is necessary to sustain and characterize MOSA and what about it is working. Such an effort is also needed to develop recommendations that will enable its success.

Until now there has not been any significant work summarizing these efforts from a holistic view. This Praxis will demonstrate just that and thus this work will be a unique and valuable contribution towards the body of knowledge and further contribute to the fields of Modularity, open systems and defense acquisition processes.

1.5 Research Objectives

The main objectives of this research will be:

- Introduce DoD's MOSA, including MOSA evolution; MOSA as defined in law, policy, and guidance.
- Analyze the impact of MOSA in the defense acquisition process and key technical aspects and standards
- Demonstrate through a qualitative approach the influence of MOSA on defense acquisition processes, aviation systems, and the emergence of similar impacts it has had on other defense programs being developed globally and commercially.
- Form recommendations for how MOSA metrics can be better measured and assessed.

1.6 Scope of Research

The extent of the research for this effort focuses on modular and open systems integration through the MOSA strategy or defense acquisition programs. The research focuses on the performance aspect of the defense acquisition programs with respect to cost and schedule. Considering there are several variables and contemporary acquisition trends such as concurrent engineering, lean and agile methods, smart manufacturing, service orientation, Model Based Systems Engineering, and autonomy that are being used to support defense acquisition programs today, more insight could be gained if there was a way to account for the independent effects of each of these variables and to compare each influence against the others. This would require more time (and information) than is available to this study. Instead, to fit the research study to its limited time span, the research was narrowed to focus only on modular and open system architecture methods.

1.7 Research Limitations

Modularity, open architecture, and MOSA are all very broad terms that can lead to various interpretations and this can limit the direct comparison of programs with respect to cost and schedule performance. The scope of this effort is broad because there is only limited information with respect to MOSA. It also turns out that there is only limited information collected. Further, there may be programs within the DoD that have already applied some level of modularity or are currently applying modular approaches using different terminology. The US DoD is the largest defense engineering organization in the world. Due to this, there are a lot of great processes, expertise, and high-level system engineers that create large scale, interdependent, and highly complex defense systems.

This creates a huge limitation as a lot of information exists between contractors, vendors, defense program leads, project managers and others, thus multiplying efforts on getting single source information about program performance. MOSA is one activity that exists to assist with building successful highly technological defense systems while trying to balance schedule and cost goals. Due to the limitations placed on US DoD programmatic information available to private citizens, not all the information needed to study MOSA's impact on programs is readily available. There might be classified MOSA program related information. Thus, it becomes tougher to regularly compare the most updated version of the project performance data for comparison.

Another limitation of this research is that it is almost impossible to attribute MOSA as the direct factor in the project performance of a defense acquisition program. Since the nature of defense programs and their acquisition cycles can be multiyear (and usually consist of multiple sub-systems), validating and gaging the impact of the MOSA strategy becomes even more challenging. Consequently, a goal of this praxis is to also shed light on how this comparison can be done better so that the positive influence of MOSA can be effectively measured.

1.8 Organization of Praxis

This Praxis is organized into five chapters. Chapter 1 is an overview of the overall research intent and will provide a brief overview of the topics and concepts discussed. The chapter further highlights the current work around MOSA is limited within the US DoD. This praxis will take this to the next step by reviewing relevant examples and effects of MOSA beyond the US DoD, evaluating MOSA on a global level, and assessing commercial aspects as well.

Chapter 2 covers the literature review that was performed as part of the research effort. The review is primarily based on peer-reviewed journals with literature focusing from government reports and papers to ensure a holistic view of the indicated gap.

Chapter 3 presents the research thesis and hypotheses and how they will be addressed through this research.

Chapter 4 contains the results of the investigation and introduces relevant case studies that will help demonstrate the effectiveness of the proposed solution.

Finally, Chapter 5 presents how this research contributes to systems acquisition and provides recommendations for future research opportunities.

Chapter 2 – Literature Review

2.1 Introduction

This review provides the results of a literature assessment for those writings that support the practical application of the research effort. This effort helped provide a better understanding of modularity, open systems, and the impact they have had on the defense acquisition process. Literature from multiple DoD, international defense, and commercial programs was integrated into the body of knowledge to ensure that there was consistent material specific to the stated problem and that the findings are applicable beyond the DoD's domain.

The overall research work further demonstrates the uniqueness of the studies performed and compiled by this praxis author. Given that swiftly evolving nature of the open and modular environment, the research of the applicable material is primarily limited to work published between 2010 to the present. There are however a few instances of some relevant material published outside of these dates if there is a direct relation to the discussion. The research conducted in the praxis is a broad literature review of DoD and other defense regulations, papers and directives related to modular and open systems to gather relevant background. Furthermore, the literature review was expanded to beyond the DoD landscape to discuss relevant strategies and effects that have been motivated from the MOSA strategy and which have had positive effects on a global level as well as in the commercial industry. For sources outside of the governmental resources, the following tools were used: Google Scholar, ProQuest Engineering Database, Engineering Village as well as other papers, briefings, articles were used to cover further content about the DoD and the

use of the MOSA strategy. The reason the literature search used a wide range of information from academia, DoD, industry, and commercial sources was purposefully done to ensure that an unbiased and balanced outlook is provided.

2.2 Modularity

Modularity can be introduced as complex systems that can be decomposed into a hierarchy of loosely coupled subsystems with standardized interfaces to assist with the design of those systems (Baldwin and Clark, 2000). While simple systems can be relatively easy to design and acquire, more complex systems can have multiple levels of granularity (i.e., the size, details, and interconnectivity of the elements). Many examples of successful modular approaches can be taken from the consumer goods, commercial sectors, automotive industries, and academia. There has been a rise in the use of modularity in sectors such as aeronautical and defense industries as well (Baldwin & Clark, 2014). Modularity has an encouraging affect with respect to new product development time and product execution, which ultimately leads to faster time to market (Danese & Filippini, 2013).

Modularity has proven to help reduce costs, accelerate innovation, and improve time to market. From computers to automobiles, there has been tremendous success with the use of a modular strategy. Using modular components, for example, Dell was able to achieve lower costs, higher sales volumes, faster delivery, and greater variety (Ro, Liker, & Fixson, 2007). Modularity has proven to increase product variation and also shorten delivery times (Ulrich, 1995). Modularity in defense military systems design has shown advantages in terms of adaptability, resilience, and cost savings (Dasch & Gorsich, 2016).

Modularity has been perceived as a strategy to better manage lifecycle costs and as well as project complexity while increasing the ability and adaptability of systems. (Heydari & Dalili, 2015).

“Common architecture, standard product portfolio elements, and a systematic approach can help companies cut costs and boost top-line growth” (Bannasch, Grüntges, Rossi, & Weig, 2018). An additional benefit is that an organization could increase competition within its suppliers, and it benefits the manufacturer to not be limited to only one supplier. Modularity has helped improve supplier integration by creating a common language to simplify communication (Lorenzi and Lello, 2001). “For the Defence sector, adopting open standards is no longer just a preference; it has become a central driver in system design.” (Clark, 2019). The ability to communicate clearly, frequently, and with lesser effort (Galvin and Morkel, 2001) has resulted in a higher level of information being exchanged and thus helping towards building a stronger relationship between the customer and supplier. In another study, Jacobs further found that modularity has led to an increase in quality with the increase in standardization of modules and led to additional benefits such as increased cost savings due to less testing and ease of technology insertion. (Jacobs, Vickery, & Droge, 2007).

2.3 Reviews of the Major Defense Acquisition Programs from 2008 - 2019

DoD acquisition programs are prone to a lot of ambiguity and risk due to the level of their complexity, organizational structure, and governmental regulations. These effects have led a lot of major defense programs to be susceptible to not meeting their initial project estimates and timelines during the acquisition cycle. As per the GAO, during their

most recent review of the Defense Acquisition Annual Assessment dated June 2020, 85 MDAPs added \$628 billion (about 54 percent) in total cost overruns and a schedule delay of more than two years (about 29 percent) as compared to their initial estimates. The report further highlighted that the implementation of software development approaches, cybersecurity practices, and vulnerability evaluations were also contributing factors to the delays. Figure 1 gives an overview of the total number of major defense programs and the total acquisition cost each year from 2008 to 2019 (GAO, 2020).

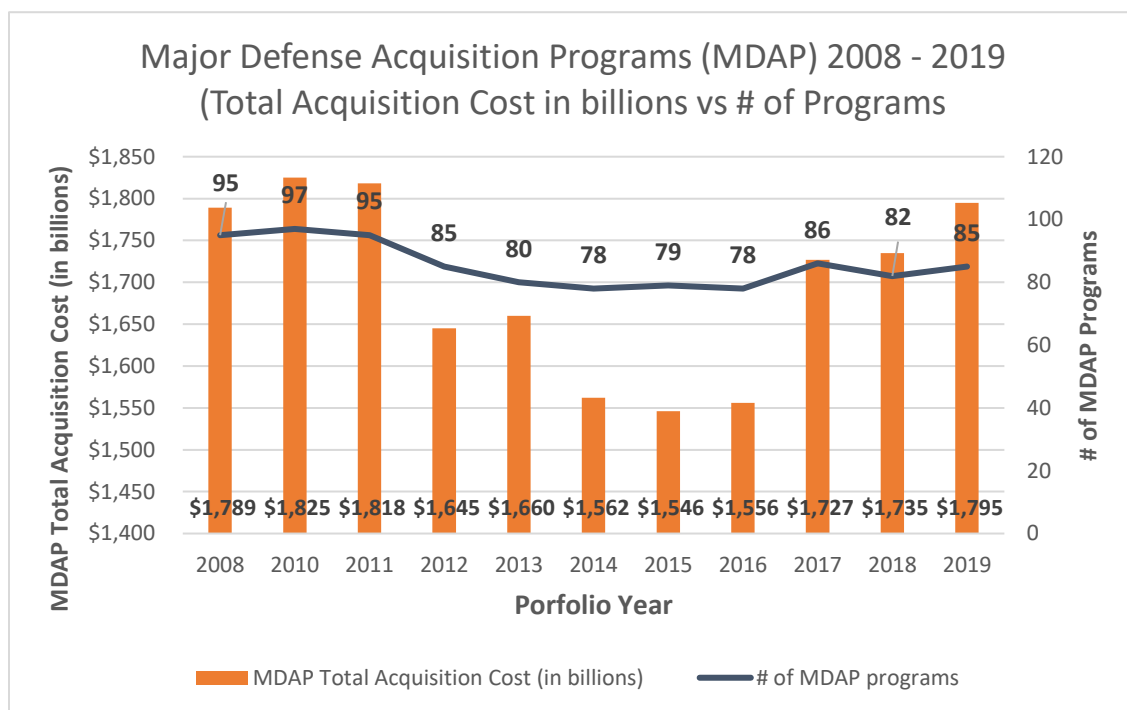


Figure 1: Major Defense Acquisition Programs 2008 – 2019

Were MOSA to have an effect, it would seem that that effect would be most obviously observable in this group of poorly performing large programs in the period right after the Department mandated MOSA (2015). Consequently, a detailed look at the MDAPs seemed promising.

2.4 Modular Open Systems Approach (MOSA)

Almost all current and future defense programs within the US Department of Defense are currently being acquired using systems engineering processes. An important change to this acquisition process is the introduction and use of an open and modular open system method known as Modular Open System Approach. The MOSA focuses on the ability to integrate modern technology and products while also ensuring the affordable and constant future upgrading of systems and sub-systems to stay ahead in the game. “MOSA can be defined as a method to design systems with highly cohesive, loosely coupled, and severable modules that DoD can compete separately and acquire from independent vendors” (Zimmerman, Ofori, Barrett, Soler, & Harriman, 2018). In fact, the DoD has taken an active interest in pursuing the MOSA methodology for its major defense acquisition programs looking through its entire life cycle activities and areas of opportunity. Analyzing this will hope to support “rapid evolution, faster cycle time, and accelerate overall delivery of defense systems” (ODASD, 2017).

MOSA is considered to be a revolutionary approach that focuses on both the business aspect but also the technical aspect of acquisition by focusing on reduction of cost when it comes to defense system growth and sustainment. DoD’s 2015 Better Buying Power 3.0 initiative focuses on the philosophy that adopting and mandating MOSA will help bring an increase in innovation and competition, in cost, and in efficiency. The Better Buying Power 1.0 (2010) and 2.0 (2013) initiatives are already seeing these benefits. “Nearly 100 percent of Acquisition Category (ACAT) I programs, approximately 90 percent of ACAT IIs, and 80 percent of ACAT IIIs now have cost targets and are managing to them, generating significant savings across the Department” (Kendall, F.,

2015). Leveraging the MOSA strategy will allow the DoD to acquire capabilities such as systems, subsystems and software that are flexible and adaptable in nature which can help respond to the changing threats and mission needs. An additional benefit is that MOSA will allow for easier technological advances, upgrades and sustain depending on program demands (Zimmerman, Ofori, Barrett, Soler, & Harriman, 2018). While MOSA is one of the many approaches that the DoD has introduced, other methods such as Lean Six Sigma, agile development, leadership changes, and system engineering tools are all factors that will play a role in its evolution. “Transformation shares success factors with other change approaches regarding sponsorship, leadership, communication, and governance. It also has additional success factors related to its approach and its enablers, both analytical and organizational” (Allaoui, Bourgault, & Pellerin, 2019).

According to OSASD, to alleviate deficiencies in the acquisition process, the DoD’s goal is that through adopting MOSA, the acquisition process will benefit in the following ways:

1. Increasing competition – By allowing components to be outsourced through different vendors through open architecture.
2. Introducing technology refresh – Periodic replacement and/or upgradation of technology on a part of the system without replacing the entire system.
3. Emphasizing innovation – Including the ability to reconfigure assets based on rapidly evolving requirements and project needs, making it more flexible.
4. Promoting cost savings – By incorporating open architecture, making components and systems more modular, reusing technology, and having the ability to increase competition among suppliers during the course of the system life cycle.

5. Improving interoperability – Ensuring that there is seamless and secure exchange of information between software and hardware interfaces, currently or in the future with respect to either access or implementation. (ODASD, 2017)

New product development coupled with innovation is indisputably vital in determining the success or failure of an organization. “Firms need to create and sustain competitive advantages to survive in today's highly competitive business environment. Timely, correct, and responsive NPD has become even more critical in the highly competitive global environment” (Fazlollahtabar & Aliahmadi, 2014). Today’s defense systems are transitioning from single components to a more networked, seamless, and connected systems of systems architecture with a goal to be more effective and intertwined. Instead of having single tanks or aircraft, it is more effective when we can synchronize multiple defense systems in one network to share communication commands and share data. The speed in innovation and technological change has increased significantly (Chao & Lenox, 2018). Software plays an essential role in MOSA in including open-source software to allow access, use, and modification for any single module with ease. Even though our defense organizations are transitioning, the current state of many programs still have single, individual, and unconnected systems that have unique software and data structures. This, along with an increase in the amount and complexity of software coding on an annual basis and having to manage the interdependence between components and systems, gives the power to vendors that have developed these historical systems to control future upgrades and maintenance and increase capabilities and reduce competition. The extent of DoD software code in service has been growing by more than an order of magnitude every decade, and a similar growth

pattern has been displayed within individual, long-lived military systems. (National Research Council, Physical Sciences, Computer Science and Telecommunications Board, & Committee for Advancing Software-Intensive Systems Producibility, 2010). The complexity of software in defense systems, specifically aerospace systems, has been multiplying rapidly. “Source lines of code in aircraft is doubling about every four years. That trend has been in place for at least five decades and applies to both commercial and to military aircraft” (Aerospace Vehicle Systems Institute, 2017). The US Army also estimated that the volume of code under Army maintenance had increased from five million to 240 million SLOC between 1980 and 2009, which represents an annual increase of approximately 15% (STODDARD, 2018). With a MOSA architecture across our defense systems, the cost of maintaining, upgrading, and increasing capabilities can largely be reduced.

2.5 Deeper Dive into Modular Open System Architecture/Approaches

Since the introduction and mandate in 2015 of the use of MOSA within DoD’s MDAPs, there has not been a detailed assessment of its use or classification. A detailed assessment would help evaluate whether MOSA has had a positive influence on the project performance on defense acquisition programs and to gauge if programs are getting better with managing cost and performance targets. One of the major concerns of the implementation of MOSA is that within the current literature, there is lack of material showing strong evidence of modularity being a direct causal variable to the success or failure of a program. This makes it tougher to gauge an understanding of how MOSA is performing with respect to its program success. “Although U.S. weapon systems are

among the best in the world, defense acquisition programs have a historical track record for being vulnerable to enormous risks in the form of cost overruns, schedule delays, and poor technical performance.”(Azizian, Mazzuchi, Sarkani, & Rico 2011).

While there is limited data showcasing cost and schedule information of MDAPs that have implemented MOSA, there is not significant literature and assessment on the implementation of MOSA. Understanding performance is crucial as it will allow the project sponsors to decide if the implemented strategy is achieving its outcomes and acquisition programs are indeed performing better in terms of cost and schedule.

Furthermore, comprehending what is working and what lessons can be learned will not only help ongoing projects but also help better support future programs as the acquisition process will keep continuously evolving. Recently during a survey provided by the United States Air Force Life Cycle Management Center, there was a study done in 2018 targeting avionic programs and the use of modularity (Falcone, 2018). In this study, there were four specific questions that gave an impression of modularity in avionic programs.

- Where do you believe your program/programs stand with respect to implementing modularity and OSA regarding both hardware and software?

Table 1: Survey Answers – Question 1

Percentage	Category
14%	Fully Implementing
33%	Partial
33%	Limited efforts
4%	Not possible
16%	Other

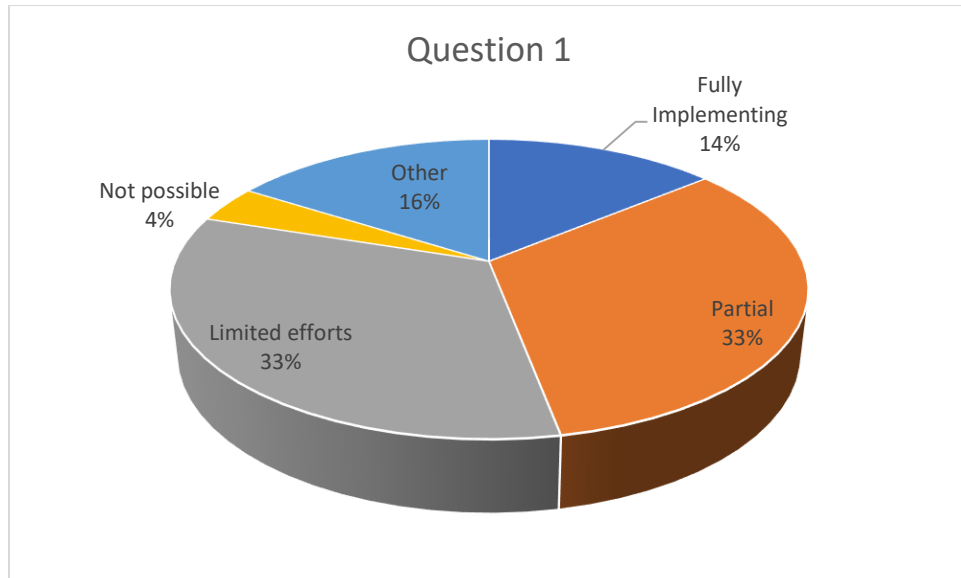


Figure 2: Pie chart for Question 1

As we can observe from the above data, that only 14% of avionic programs are fully applying MOSA into their programs while almost 66% represent only partial or limited implementation.

- Where do you believe your program/programs stand with respect to implementing modularity and OSA for new development or both?

Table 2: Survey Answers - Question 2

Percentage	Category
26%	Fully Implementing
43%	Partial
17%	Limited efforts
3%	Not possible
11%	Other

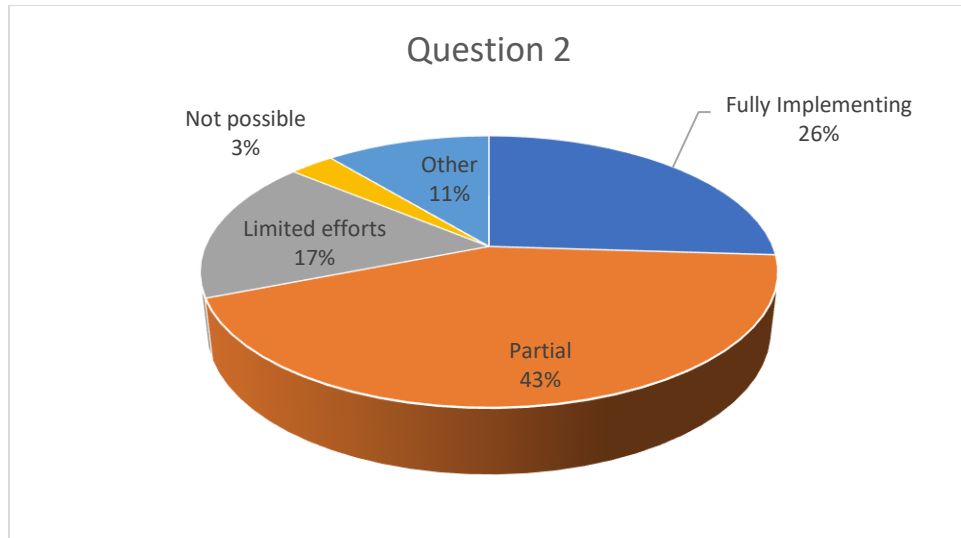


Figure 3: Pie chart for Question 2

As we can see from the second chart above, when it comes to newer development, almost 70% of the programs are implementing MOSA either fully or partially during the acquisition process.

- What are the challenges/barriers to implementing more modular or open system architectures?

Table 3: Survey Answer – Question 3

Percentage	Category
15%	Lack Ref. arch or tools
15%	Lack Govt skill sets
34%	IP/ industry resistance
15%	Not a priority
21%	Other

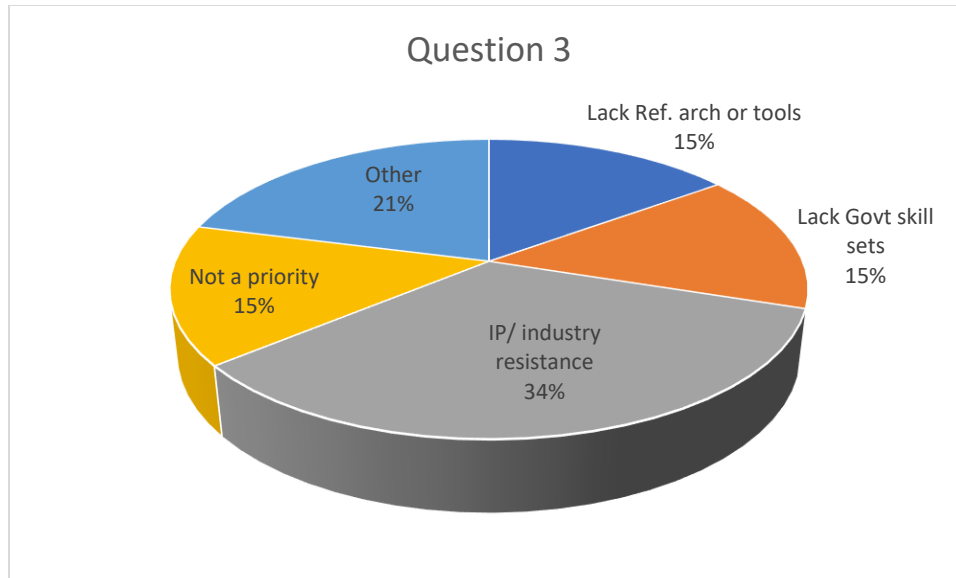


Figure 4: Pie chart for Question 3

A third of the respondents cited Intellectual Property and/or industry resistance as one of their top challenges when it comes to implementing modularity.

- Do you have the tools you need to assess where your system/systems stand regarding modularity or OSA? (if yes, please list what assessment tools you're using)

Table 4: Survey Answer – Question 4

Percentage	Category
25%	Yes
69%	No
3%	NA
2%	Limited
1%	I don't know

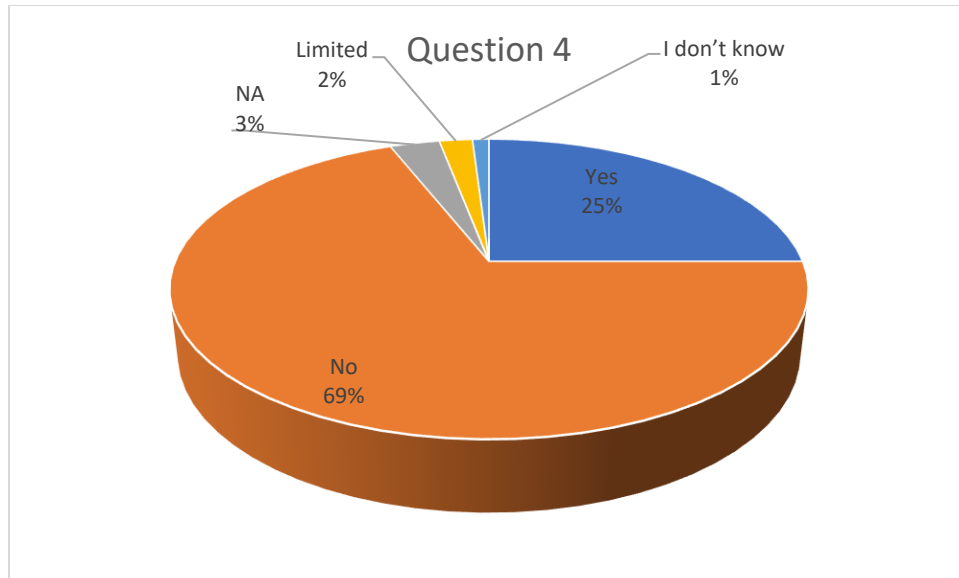


Figure 5: Pie chart for Question 4

From the above chart, it is very clear that there is a strong need for an assessment methodology and tools to better gauge the understanding and impacts modularity has on the programs. In fact, even though the DoD has mandated assessment of technology maturity for its major defense acquisition programs, there has been little to no research on the quality of the acquired defense systems or their performance.(Azizian, Mazzuchi, Sarkani, & Rico 2011)

2.6 Advantages & Roadblocks with Modularity and Open Approaches

As the larger defense community looks to accelerate the integration of modularity and open architecture, it is vital to discuss the current advantages and barriers associated with open architecture and modularity.

- **Managing complexity and increased innovation**

Today's systems are getting increasingly complex in size, interaction, components, systems, and external factors. Since modularity has seen success in other commercial

fields, there is an opportunity to harness it in the defense acquisition process. One of the sectors that has seen considerable development has been the automotive sector. From the late 1990s to the early twenty first century, the automobile sector has made huge strides in designing standard platforms within automobile manufacturing. The transition to standardization of platforms has led to more efficient design development and execution. That in turn has assisted these organizations to release multiple variations and models of automobiles with ease, targeting different market segments (Lampón, Cabanelas, Benito, 2017). An added advantage of modularity is that it helps transition old products and allows them to refresh with variation in functionality through the interchanging of modules. (Gu and Sosale, 1999).

- **Increased flexibility and adaptability**

“One of the key advantages of an open architecture is that it allows the design to be flexible and adaptable.” (Engel & Browning , 2006). “Modularity is one of the most useful tools employed in the product development process. Regarding functionality, the use of modules is common method to generate flexible platforms to manufacture products and product families that require functional variations” (Mesa, Esparragoza, & Maury, 2019). Employing this strategy, a standard design for a system could be used by the Army, Navy, and the Air Force with minimal modifications. In order to keep up with ever-changing technology and customer requirements, an adaptable design is required. “The underlying philosophy is to adapt to changing environments while reusing the design or parts of the design” (Gu, Hashemian, and Nee 2004). From a product design viewpoint, the recent modular platforms have increased the number of models that can be included in comparison with standard platforms. This is due to the fact that “modular

platforms allow for different dimensions in this structural element and the assembly of models from different segments” (Buiga 2012). The added flexibility that modularity brings will allow systems to quickly adopt changes in design, quantities, and technological advances all to be incorporated swiftly rather than going back to the starting point. Another advantage is that due to the high level of complexity, systems and sub-systems can often be designed by different teams and vendors, allowing the project to not only be worked by different groups but also concurrently and yet be integrated into one single system. “This approach has led to significant cost savings and spurred new innovation in the automobile industry where all major automobile manufactures have developed their own modular platforms such as Volkswagen’s MQB (Modularer Querbaukasten) platform, PSA Peugeot-Citroën’s EMP2 (Efficient Modular Platform 2), Renault-Nissan’s CMF (Common Module Family), BMW’s UKL (Unter Klasse) platform, Daimler’s MRA (Mercedes Rear-wheel-drive Architecture) platform, General Motors’ D2XX (Delta 2 XX) platform, and Volvo’s SPA (Scalable Platform Architecture)” (Lampón, Cabanelas, Benito, 2017), all of which have led to cost savings, an increased variety of automobiles, and faster time to market and has led to radical advances in modular approaches.

- **Fights product obsolescence, reduces new product development time, and increases overall efficiency**

As technology advances, so has obsolescence. “For long lifecycle systems such as defense systems, it has become a major concern as the lifecycles of the components that are encompassed within these systems are often far shorter - up to ten times shorter - than the overall system lifecycle.” (Devereaux, 2010).

To ensure weapon system acquisitions remain state-of-the-art, it is essential to build systems that meet current needs yet also evolve to address future uncertain needs. One crucial advantage of open or modular systems are, that they allow us to expand innovation and meet the evolving customer's changing requirements. Modularity can help mitigate the risk of market loss and technological obsolescence. Having products separated into product families has helped balance high variety as well as manage resources optimally (Gonzalez-Zugasti, Otto, and Baker, 1999). Some of the most successful examples of modularity which we can see are in our everyday lives, the computers we use often come from product families or modules. Our mobile phones have 80% similar components with 20% adjusted features that are targeted to different customers and target markets. For example, an iPhone XS is a premium product that targets a niche group of people that value high quality phones and are willing to pay a premium price whereas an iPhone XR is more focused on consumers who want an Apple phone with a moderate price range. Another example of modularity is computers, where one can order a computer online and personalize the size of screen, the CPU performance, and the computer's looks based on one's needs and wants. In the example of both the laptop and iPhone cases, the components are 80% similar and features and price will vary based on consumer needs and wants.

Clothes, bedding, and furniture can be mixed and matched from different retailers and manufacturers. Even though equipment comes from different manufacturers, there are standardized requirements and sizes that allow more companies to innovate and compete in the marketplace. "Modularity does more than accelerate the pace of change or heighten competitive pressures. It also transforms relations among companies. Module designers

rapidly move in and out of joint ventures, technology alliances, subcontracts, employment agreements, and financial arrangements as they compete in a relentless race to innovate” (Baldwin & Clark, 2014).

As discussed above, modularity has some great benefits and advantages. The primary driver for the implementation of MOSA in new acquisitions by January 2019 comes from the congressional mandate in the 2017 National Defense Authorization Act. Since MOSA is fairly new in the defense acquisition marketplace, there are still a lot of gaps and opportunities to focus and improve its application. Because the defense industry often develops low volume, high-cost systems, it is essential to define MOSA and modularity where it can be synchronized and used directly. The rules and standards will often vary for simpler systems as compared to assessing modularity in more complex defense systems. Some of the biggest areas to address MOSA are in system development activities where there is a:

- **Lack of a MOSA framework** - A critical area where there is lack of an effort is a solid framework for supporting the development and the integration of modular or open architecture in forms of specific methods and standards in the defense industry as compared to the robust usage of modular architecture in the consumer goods and automotive industries. “MOSA must become commonplace, with design patterns, instantiations, tool use, measurement, etc., ingrained in the way defense programs conceive, design, build, and maintain systems” (Zimmerman, Ofori, Barrett, Soler, & Harriman, 2018).
- **Absence of MOSA metrics** – While there are metrics related to modular architecture in general, there is a lack of specific modularity metrics for acquisition within the

DoD. This makes it hard to quantify modularity in terms of return on investment and cost savings associated with modularity. There is already an awareness of the potential cost saving benefits of modular and open architecture, yet there is no specific way to measure direct impact and savings with respect to cost or schedule due to the implementation of MOSA.

- **Lack of incentives** – While MOSA has been mandated by the DoD, currently there are no incentives for program managers or vendors to support modular initiatives. There are also no negative consequences for the failure to use modular or open architecture.
- **Lack of formal assessment and evaluation tool** – Currently, programs that use MOSA are not mandated to use DoD's evaluation tool known as the Program Assessment and Rating Tool. As per the Government Accountability Office (GAO), the Program Assessment and Rating Tool is an optional questionnaire that evaluates replies based on a set of questions responded to by DoD program managers to assess how modular a program is as per them. The questionnaire is evaluated by answering each question from four options – None, Little Extent, Moderate Extent or Large Extent. These answers are then computed using an algorithm and a two-dimensional score is calculated that gives a visual chart of how open the corresponding system being evaluated is. A score ranging anywhere from 0 which would imply the systems is not very open to a score of 4 which would indicate that the system has enterprise openness. This score gives the user an impression of the modularity in the system. While this is a start, a concern with this tool is that it is biased since the person using the tool can be easily varied from one member of a team to another. It is also not a

mandatory tool and can thus be hard to compare between different programs. “What to measure and how to measure the MOSA implementation to ensure that MOSA is being used to the largest extent practicable is one challenge that will be difficult given the combination of elements that contribute to a successful system outcome.”

(Zimmerman, Ofori, Barrett, Soler, & Harriman, 2018).

One DoD example of modularity that is of particular interest is the comparison of the B-52 Stratofortress versus the B-58 Hustler. The B-52 was built by the Boeing Military Airplane company and flew its first mission in 1954. For over six decades, the B-52 Stratofortress has been a shining star when it comes to manned strategic bombers within the United States’ defense strategy. “The B-52 is known for its versatility of dropping or launching a variety of weapon systems such as gravity bombs, cluster bombs, joint direct attack munitions, cruise missiles, nuclear weapons, and precision guided missiles. Current engineering analyses show the B-52’s life span to extend beyond the year 2040” (Military.com, 2018). The flexibility and adaptability of the B-52 makes it a very valuable defense system and if modernized with latest technology, the B-52 will remain as a strong contender in securing the country’s defenses. While this was not a planned modularity project, the ability to equip it with new technologies has made it one of the most modular defense systems in existence and has also saved it from becoming obsolete. In contrast, Convair’s B-58 Hustler which was one of the most exciting projects during the Cold War was obsolete even before its first mission in 1956 and retired within a decade. In fact, the B-58 was more complicated, much harder to design, thrice as costly as the B-52 program. Moreover, the project to reengineer the fuselage into the “Coke bottle shape” called for by area rule aerodynamic theory led to further delays in schedule

and cost, causing a catastrophic end to the program (Beckhusen, 2017). The Hustler, while a supersonic aircraft, had lots of constraints and limitations that did not allow it to change with the times and thus was an expensive mistake that became a valuable lesson for engineers around the world. Even after a period of sixty years of service, the B-52 continues to adapt to a variety of missions such as supporting close air support and open ocean reconnaissance. The Air Force intended to continue to leverage the modularity of the B-52s and extend its service up to the year 2045, bringing the B-52 to being continuously in service for over 90 years. Even though the B-52 was not developed to be an example of a modular aircraft, the program remains the most modular and effective program serving multiple mission requirements within the avionics program.

2.7 Adoption of ISO/IEC/IEEE 15288 by the DoD to increase interoperability

The move towards better acquisition reforms by the DoD has led to more non-governmental and widely acceptable open standards. The DoD acquisition process requires programs to use MOSA and react more quickly to technology, threats, and mission changes. This has further pushed to provide more interoperability between standards. The adoption of the ISO/IEC/IEEE 15288 standard, Systems and Software Engineering – System Life Cycle Processes was implemented by the DoD. This step has allowed the acquisition process to be more streamlined and catered to a more systems engineering related standard thus terminating numerous existing military specifications and standards. The ISO/IEC/IEEE 15288 allows for a universal framework for system lifecycles and focuses on a concentrated set of systems engineering processes that are used in the concept, expansion, manufacturing, application, support, and retirement lifecycle stages (Daly 2017). “Pat Shanahan, the then Deputy Secretary of Defense

overseeing the reorganization of the former chief weapon buyer into two new positions, said speed and rapid enhancement of existing systems with emerging technologies is a personal goal.” (Parsons 2017). In fact, two new DoD System Engineering Non-Government Standards were established by the DoD complementing ISO/IEC/IEEE 15288:

1. IEEE 15288.1 - IEEE Standard for Application of Systems Engineering on Defense Programs
2. IEEE 15288.2 - IEEE Standard for Technical Reviews and Audits on Defense Programs

Both these standards were published on May 15, 2015 and first used within the DoD the following month on June 5, 2015. Both these standards were developed to specifically provide guidance on definition, control, and improvement for DoD acquisition projects with an SE focus. “The 15288 Standards provide one method to define the acquirer’s expectations and requirements for the supplier’s performance of SE processes and technical reviews and audits. Thoughtful and proper use of these standards can enhance communication and understanding between the acquirer and supplier throughout the solicitation process and contract execution.” (DoD Best Practices, 2017). This ties directly to one of the five benefits of MOSA set by the DoD which is increased interoperability for mission integration. Furthermore, INCOSE in its 2015 Systems Engineering Handbook – which is based on ISO/IEC/IEEE 15288 - lists modularity as key to establishing an agile system engineering framework. (INCOSE 2015, Ref p. 208-210).

2.8 An analysis of Modular Systems Approach in Avionic Defense Systems

There are several programs currently ongoing with the DoD that are focusing on implementing open system architectures. One field where the US military is aggressively pursuing open systems is Avionic systems. Some major programs that are implementing Open System Architecture in the avionics industry are the “RQ-4 Global Hawk and MQ-9 Reaper unmanned aerial vehicles; AH-64 E Apache, and HH-60W Black Hawk Rescue Helicopters; U-2 Dragon Lady; F-22 Raptor; and the AV-8B Harrier jets” (Tokar, 2017). “Modularity must be a complexity management mechanism that incorporates these novel schemes such that system architectures can take advantage of them within complexity management in future generations of engineering systems.” (Heydari, Mosleh, & Dalili 2016).

2.9 Global effects of MOSA

The emergence of the adaptable open system approach of MOSA has led to many studies and partnerships at a global level and in the commercial sector (Welby, 2014). The UK Ministry of Defence (MOD), Australian DoD, & the Canadian Armed Forces (CAF) and commercial partners such as General Electric, Boeing, Airbus, and others are adapting the ideas of modularity and incrementally developing their systems to be aligned with the MOSA principles. Collaborative approaches are being adopted and undertaken by the Australian DoD, US DoD, and UK Ministry of Defence, sponsoring studies in partnership with both industry and academia; much of this work advocating for the benefits of more open technical and business practices (Ollett, Coleman 2015). Due to this, there has been a rise in the number of projects and MOSA initiatives with respect to procurement and development of defense systems, all with a common theme of

leveraging technical, cost, and project benefits specific to adaptable and modular systems.

2.10 Literature Review Summary

DoD acquisition programs are prone to a lot of risk and uncertainty because of the intricacy of the defense systems, constant change in program needs and requirements and coordination between government and commercial organizations. These consequences lead to a higher risk to program cost and schedule adherences. In response a large percentage of major defense acquisition programs being delayed or over budget, the DoD has formally mandated MOSA as a strategy to enhance the DoD's ability to acquire and modify defense systems competently. Given the complexity of acquiring defense systems, it is important to understand the mandate of the MOSA strategy and the specific variables that are associated with the process. The DoD's vision for MOSA is not only a technical shift in strategy but also a huge cultural change with respect to ways of working, law and policy. Since the applied research effort is intended to assess a specific DoD MOSA strategy, the literature review, fundamentally includes a significant number of DoD references, technical standards, manuals, papers, presentations, and specific studies of modularity and open systems. Several sources outside of the DoD literature are also included to ensure comprehensiveness, thoroughness, and a holistic review.

From the literature review there is a good understanding that MOSA, modularity, and open systems are all used in similar ways and have often been defined differently depending on the business unit. With global defense spending increasing every year, there is an increased burden to make certain that major defense program acquisitions are efficient and able to achieve its cost and schedule targets. There is a gap on how MOSA

is being assessed and how the benefits are being balanced. What to measure and how to measure modularity's impact remain key challenges. The next chapter will focus on introducing hypotheses and how they were assessed for this research.

Chapter 3 – Methodology

3.1 Introduction

Chapter 3 presents and describes the chosen research methodology. The chapter focuses on introducing the hypotheses and how they are assessed for this research.

3.2 Hypothesis

The six Research Hypotheses connected with this research effort are:

H1: If MOSA is successful, then the expectation is to see good cost and schedule effects available from program performance stats attributable to modularity.

H2: If MOSA is successful, then there is evidence that government programs will claim so and not claim otherwise.

H3: If MOSA is successful, then there is evidence that other US and non-US government departments will also require it.

H4: If MOSA is successful, then other US departments and non-US government departments will show evidence of its success and not claim otherwise.

H5: If MOSA is successful, then there is evidence that the commercial sector has/is adopting it.

H6: If MOSA is successful then there is evidence that the commercial sector is having success with it and not claim otherwise.

3.3 Hypothesis Overview – H1 - Cost and Schedule Effects

In order to evaluate possible effects of MOSA on DoD programs, this praxis looked at the 16th Annual Assessment report of the DoD's MDAPs released by the GAO. "The

GAO is an independent, nonpartisan agency that works for Congress. Often called the congressional watchdog, GAO examines how taxpayer dollars are spent and provides Congress and federal agencies with objective, reliable information to help the government save money and work more efficiently.” (Government Accountability Office, 2018).

The GAO’s annual assessment report includes performance, cost, and schedule evaluations of DoD’s 86 MDAPs from the portfolio of FY2017. This report was released on April 25, 2018. The total value of the 86 programs in the report equaled \$1.66 trillion and detailed assessments were made in the report with respect to cost, schedule, and quantities of items developed. Every MDAP is also assessed through a document called a Selected Acquisition Report which looks at performance markers such as initial cost, schedule, and performance baselines and compares this to the current progress of a system’s development in its acquisition life cycle.

MOSA had been introduced by the DoD as the Department was under great pressure to find smart ways to reduce costs, focus on increasing innovation, and improve quality. MOSA was mandated under the Better Buying Power 3.0 Guidance that focused on “Achieving Dominant Capabilities through Technical Excellence and Innovation,” while also continuing efforts started with the Better Buying Power 1.0. The praxis research focused initially on MDAPs because a fair amount of information was available on these programs and combined with information from the Selected Acquisition Reports, it was reasonable to conclude that the effects of the introduction of MOSA could be detected and assessed from this cost and schedule information.

The 86 MDAP programs’ assessment reports could potentially identify whether

MOSA has contributed towards the DoD's goals to meet cost and schedule targets while also potentially seeing if the introduction of MOSA had led to faster innovation and product development. Since defense programs have high levels of regulations and classified matter, to protect national interests, it was difficult to find sufficient information on the performance of defense programs. But with the GAO's annual report on the evaluation of the major defense acquisition programs and an accompanied Selected Acquisition Report, taxpayers can gain a deeper understanding of defense acquisition programs and their performance. Studying the MDAP programs in detail has the further advantage that it allows the development of further clarity on whether MOSA is making a direct impact to project performance in the acquisition process.

Understanding this will further clear up whether focusing on MOSA was the right call by the DoD to address faster innovation and in assisting it to meet cost and schedule targets.

The DoD acquires its weapon systems through a process called the Defense Acquisition System that has three key phases. (Department of Defense Directive 5000.01). The three major phases are:

1. Technology Maturation (Technology Development)
2. Engineering & Manufacturing Development (Systems Development)
3. Production & Deployment (Production)

The praxis focused on understanding all these three above phases and working towards extracting data that is publicly available with respect to the defense programs. Data such as cost, schedule, and increase or decrease in quantities are all captured annually in the form of latest estimates through the Selected Acquisition Reports. Programs that demonstrate a unit cost increase of fifteen percent or more and programs

that carry a schedule delay of at least six months are reviewed in depth to better understand the variances.

Modular open systems could be of great value to the acquisition of defense systems. As systems grow more complex, traditional approaches to promote innovation will no longer be able to support the needs and requirements for current and future needs. “The technological products and systems that we design and use today are characterized by more and more components and increasingly intricate interdependences, implying increasing design complexity and challenges for future innovation” (Luo, 2017). One of the key focus areas of introducing MOSA is to help work towards solving growing complexity and introducing innovative practices. Because the amount of investment in defense systems is very high, it is essential to properly utilize modular strategies and benefits so that we can achieve the goals set by the DoD on MOSA. MOSA is already benefitting a few current and legacy programs.

As the results in Chapter 4 will show, the findings of this phase of the research, while very interesting, were ultimately insufficient to statistically support the original hypotheses and fully isolate the MOSA effect from its confounding variables. As a result, the research then pursued a search looking through a qualitative lens.

3.4 Hypothesis Overview – H2 – MOSA as a successful strategy

The second hypothesis focused on qualitative evidence as well as successful examples furthering the case that MOSA is an effective strategy within the government programs it bene mandated. The reason for the implementation of MOSA was for the DoD to react to the slipping cost and schedule targets and develop a revolutionary strategy that would focus on the business aspect as well as the technical aspect of

acquisition of defense systems with an emphasis on reduction of cost and schedule parameters when it comes to defense system growth and sustainment. DoD formalized this move through the 2015 directive of the Better Buying Power 3.0 mandate of incorporating MOSA framework in all its projects. A few other initiatives such as the National Defense Act in 2017 and the adoption of the ISO/IEC/IEEE 15288 focusing on increasing interoperability further highlighted the MOSA strategy is here to stay and is part of the long-term defense acquisition planning tactic.

Chapter 4 will further discuss the implementation of the MOSA strategy in the acquisition projects within the Army, Navy, and Air Force. There will be a further deep dive on examples where MOSA has been applied to specific projects and there has been some great success stories and lessons learned that have come out from that experience.

3.5 Hypothesis Overview – H3 & H4 – US and non-US Government department impacts

The next set of hypotheses will focus on two specific impacts. The first focus will be with respect to the different DoD departments such as the US Army, Navy, and Air Force and how these departments have fared with the MOSA strategy. One area within these departments that has had a lot of success is in the avionics sector and specific strategies around modularity and open systems will be assessed. The second focus will be the impact of the MOSA strategy with respect to non-US governmental departments and other global defense governments such as the U.K. Ministry of Defense, Australian DoD, and the Canadian Armed Forces that have each taken similar efforts on acquiring adaptable and modular defense systems after reviewing the accomplishments the MOSA

strategy has had within the US DoD.

As discussed in the literature assessment, the DoD avionics community has had notable successes with modularity. There are currently numerous efforts ongoing to address the instructions specified in Better Buying Power 3.0 and other government instructions with respect to modularity and open systems. Consequently, the praxis research was focused on three major successful DoD avionic programs to understand the MOSA effect better:

- The Unmanned Systems Command and Control (UCS) Architecture which is mainly interested focuses on the control segment of unmanned systems
- The Open Mission Systems (OMS) Initiative which highlights efforts on new airborne and legacy platforms
- The Future Airborne Capability Environment (FACE) Methodology which is working towards a common operating environment for all aerial systems.

From these programs, the praxis research attempted to gain a better understanding of how the mandate of MOSA on defense acquisitions programs is being executed and impacted. The investigation of DoD avionic systems was fruitful in that it shed new light on what was working with MOSA and its challenges, and that it provided insights towards the measurement of its effects. These findings will be reviewed in the next chapter. Avionic systems have seen a great deal of success in their programs with respect to the push towards building defense systems that are more modular and open in nature. While the DoD avionic program findings were insightful, it was not clear whether the insights (and the challenges) the literature search revealed were limited to just the DoD avionic programs or whether there were other programs within the DoD that had seen

similar learning curves and achievements. Consequently, the praxis research assessed how other nations and the commercial sector have taken on the MOSA challenge and how they have fared. Revealing the success of modularity in the acquisition efforts of other defense departments found in the earlier chapters of the praxis, suggested that vendors could perhaps see first-hand positive impacts as well as transfer the technology to their commercial offerings. The praxis research was then directed to see if MOSA was executed differently in non-DoD areas such commercial organizations and global defense organizations, and if the effects were being measured differently.

3.6 Hypothesis Overview – H5 & H6 – MOSA and its impact on the commercial sector

Since DoD had seen initial success with modularity to the point of mandating it, it was reasonable then to conclude that perhaps other defense departments might have followed DoD's lead and mandated modularity or had incorporated modular approaches like MOSA into their acquisition strategies. The praxis research pivoted to investigate how and where this was occurring. It turned out that other defense sectors within the DoD such as the US Navy and US Army have seen good effects from adopting modularity and also working with their vendors on aligning efforts. These findings will be reviewed in the next chapter.

In general, defense spending across the globe is transforming its strategy to be more focused on flexible, responsive, and increased vendor competition for its current and future defense programs. From life cycle support to a stronger focus on Commercial Off The Shelf (COTS) wherever possible. Defense organizations around the world are under

more scrutiny and increased pressure to deliver defense systems on time and on budget as public funds are used. The environment is changing, and national security remains a priority for governments as they build and preserve defense capabilities.

As the following chapters will show, the chosen praxis research methodology was useful in that it established broader patterns and located specific examples of MOSA's effect on large developmental programs. The research approach also shed light on MOSA's extensibility so additional research was conducted to better understand the effects and impact of modularity in defense systems.

The US DoD has been applying pressure to its vendors and contractors to deliver its defense systems on time. This has led increased pressure on defense contractors and their supply chains to further deliver to governmental expectations. It would be logical to conclude that vendors being asked to deliver modular systems to their defense clients would not only learn effective modular approaches but would also witness its benefits firsthand and thus would be likely to apply similar business practices to their own products if they judged them beneficial. The literature reveals this to be true. There have been quite a few success stories in the commercial sector; this is discussed in greater detail in the following chapter within section 4.4.

3.7 Summary and Conclusion

Diving deeper into programs within the DoD gave additional insight on how the newer programs were adopting open system approaches. Different approaches and strategies were observed within the branches in the DoD. While some programs were limiting modularity's scope to sub-systems and software synchronization such as the

Consolidated Afloat Network and Enterprise Services plan, some programs were adopting MOSA strategies at all levels including having a common platform similar to the US Navy's effort with the Multi Mission Platform. While reviewing the limited quantitative analysis, there was little to no information on how programs using the MOSA strategy were being evaluated on ongoing defense procurement projects within the US DoD. It seemed to be impossible to isolate the effect of MOSA from other possible confounders and due to this, the research shifted to looking for qualitative evidence to answer the questions the methodology posted. During the qualitative analysis, more questions arose on how MOSA was being measured and to what detail. Further investigation led to assessment through the MOSA Program Assessment and Rating Tool. In fact, this assessment tool is an adaption of an existing assessment tool called as the 'OMB Program Assessment and Rating Tool'. This tool was made available for evaluating the openness of a defense system primarily managed by the self-evaluation across a few parameters answered by program managers and sponsors. However, the assessment tool was designed in a way where the use of the tool was completely optional, and the questions were answered through a survey format by respondents managing the programs that could skew the results and fail to provide a complete impartial picture. A detailed discussion about the tool was provided in chapter 2. Because of all these factors, understanding and assessing MOSA programs was critical and thus the research shifted its focus to more of a qualitative study. These insights, discussed in Chapter 5, resulted in meaningful guidance to programs for how to close the important MOSA measurement and assessment gap needed to keep this valuable acquisition approach moving forward.

Chapter 4 – Results

4.1 Introduction

Chapter 4 contains the results from the methodology presented in Chapter 3. Chapter 4 highlights a review of the six hypotheses discussed in the earlier sections touching on MOSA as a successful policy, its impact in the various government departments, its impact on other global defense programs and its effect on the commercial side of things. The chapter will discuss patterns and practices found in each hypothesis in detail. The chapter concludes with a summarized review of these results.

4.2 Hypothesis Overview – H1 – Cost and Schedule Effects

For this hypothesis, a data set was reviewed from the 2017 GAO report published by the DoD. In this report, there were 86 MDAP programs that were carefully reviewed and classified if they had modular aspect to their acquisition process or not. These programs were then compared to better understand a relationship to see if programs that used the MOSA strategy had a greater project success as compared to programs that did not implement the MOSA strategy. A Two Sample T Test was used to determine if both populations would lead to some convincing results. There were two data sets that were reviewed for this analysis.

Data Set I – This data set included a collection of all the 86 major defense acquisition programs in terms of the total cost spent on the program to date and an overview of how the programs performed with respect to the initial planned cost versus actual cost. More specifically, the research compared the cost of the first full approximations of the overall acquisition costs (in financial year dollars) from the DoD's 2017 MDAP list. Table 1 in

the Appendix lists the cost comparison along with a percentage change from initial cost to current cost. From the 86 MDAP programs, there were 66 programs that used or are currently implementing modular or open architecture methods during the life cycle process of the program. The remaining 20 programs showed no use of modular techniques. This lack of MOSA adherence could be because of factors such as the programs were upgrading or increased quantity re-orders and there was no requirement or justification to apply modularity practices to such programs. Resulting from this investigation, the research observed that of the 66 MDAP modular programs, 46 programs were over budget with respect to cost from their initial estimate, and 20 programs were within or on their initial budget. Similarly, of the 20 MDAP Non-modular programs, the research found nine programs that were over budget and 11 that were within their initial estimate of their program budget.

Data Set II – In this dataset, the research looked at 57 programs within the 2017 MDAP list which specifically had detailed one-page summaries in the GAO 2017 Annual Weapon System report. The reason these fifty-seven programs were chosen is because at the time of the report, 45 of the 57 MDAP's were between the beginning of the developmental cycle and between the preliminary stages of production. The remaining 12 were future MDAP programs which may later enhance our understanding of acquisition reforms. All these 57 programs align with the Better Buying Power 3.0 Interim Release which focuses on addressing sound management practices, the use of prototypes, more realistic cost estimate measures, and the use of sound systems engineering principles. Of the 57 present and future programs that were evaluated, 55 programs stated that they intended to encourage competition between vendors throughout the lifecycle of the

system acquisition. The remaining two future programs emphasized that their current project plans did not involve or prioritize competition either before or after development started. Program offices report they use the following measures to promote competition:

- Use of modular open architectures, that promotes the use of several modules which work together as a cohesive system that can be accomplished as a separate sub-system and also be acquired independently from the same or different vendors, thus promoting competition and parallel development.
- Along with the acquisition of the specific system, to also acquire the related system specifications and drawings as part of the technical package from the vendor. (GAO, 2017).
- “The two programs that have no current plans to use modularity are the Air Force’s B-2 Extremely High Frequency Satellite Communications system and the UH-1N Utility Helicopter Replacement program which may or may not change its priorities during the lifecycle process as they evolve” (GAO, 2017).

From the 57 programs with respect to schedule, 23 programs have not yet started and thus a schedule comparison analysis is not possible. When the research looked at schedule, out of the 34 programs, 27 programs were found to be over schedule. This indicates that about 79% of programs are already delayed. Similarly, while looking at cost, out of 34 programs, 19 programs are over budget which equates to about 56% of the programs.

The findings of this phase of the research, while very interesting, were ultimately insufficient to statistically support the hypotheses and fully isolate the MOSA effect from its confounding variables. Strategies such as the application of lean, agile, six sigma,

commonality, specific systems engineering methods, model-based systems engineering initiatives, or even a change of leadership over the term of the project could be contributing confounders to the success or failure of a programs.

“Modularity’s benefits depend on the goal of the modularization and the resultant mode of modularity. They are difficult to assess, let alone measure quantitatively. This presents a challenge to decision makers, particularly in the defense industry, where extremely expensive systems are purchased with taxpayer dollars and expected to perform for decades in some of the most dynamic environments” (Wilson, 2017).

Overall, the results show that currently it is impossible with the published MDAP performance data alone to isolate the effect of MOSA on defense acquisition programs and measure its impact. Since defense acquisition programs are multiyear and there are so many confounding variables associated with large scale projects like these, it makes it even more challenging to evaluate the efficiency of the MOSA strategy. An additional reason for this is because the sample size is too low and since a lot of the newer programs are still in the developmental phase, there is no data currently available. This further suggests that there is a strong need for better metrics to measure performance of MOSA within acquisition programs as well as more programs with a standardized assessment tool to measure and compare all DoD acquisition programs across the organization. As a result, the research then pursued a qualitative based search for additional evidence which will be further discussed in the next stages.

4.3 Hypothesis Overview – H2 – MOSA as a successful strategy

The hypothesis mainly focuses on whether there is sufficient qualitative evidence

showcasing that MOSA has been a successful strategy so far within the DoD acquisition process. Below we will go through a few examples where the MOSA strategy has made some great strides in the acquisition process and achieving its intended outcomes as well as a few examples where trying to incorporate multiple needs led to scope creep and eventually resulted in a massive impact to the program in terms of cost and schedule delays.

In the Navy, the Consolidated Afloat Network Enterprise Service (CANES) has been a successful project where the CANES Network has not only brought more magnitude and technology, but also a project that will meet requirements in cost and schedule. (SPAWAR, 2011). “CANES is the consolidation and enhancement of the requirements for five existing legacy network programs, as well as a single support framework for all (naval) C4I applications that currently require dedicated infrastructure to operate delivered and managed legacy systems. CANES will take advantage of the new business model of open architecture, Service Oriented Architecture (SOA), and rapid COTS insertion, in order to bring fiscal savings to the Navy, as well as operational agility to the warfighter” (US NAVY, 2018). The Army’s Future Airborne Capability Environment that integrates software components from different vendors under a common architecture and the Air Force’s use of Open Mission System which is an open architectural standard focusing on simplification of mission systems and increasing technological refresh are some strides in the goal to achieve modularity and its objectives. While these are a few of the success stories, some projects such as the F-35 Joint Strike Fighter, tried to use MOSA to accommodate a lot of requirements such as building a universal aircraft that can support multiple support missions for the Army, Navy, and the Air Force. In

accommodating multiple requirements, the program encountered major delays and issues. “The F-35 program is expected to cost more than \$1.5 trillion over the course of its 55-year lifespan with huge delays in schedules and high overbudget issues” (McCarthy, 2018). This is a prime example where MOSA principles have been used but the defining of boundaries and requirements has led to the use of a MOSA strategy that appeared to fail raising the question of whether MOSA as an approach failed or whether it was poorly applied. To avoid future MOSA program failures there is an urgent need to capture lessons learned from these failures and have a better evaluation tool that will help with understanding the balance between the use of MOSA and also achieving project success in terms of schedule and cost.

Some further successes that have come from the MOSA strategy. One key program discussed below is the Consolidated Afloat Network Enterprise Service (CANES).

Consolidated Afloat Network Enterprise Service (CANES) – The CANES program is the Navy’s future generational tactical network which embodies the MOSA strategy. In this dynamic strategy, CANES will have the ability to consolidate and improve up to five legacy shipboard network programs to allow for a common computing environment for over forty command, control, intelligence, and logistic applications. This further will provide a platform that is agile, flexible, and responsive to the different warfighter environments. “This strategy strengthens the network’s infrastructure, improves security, reduces the existing hardware footprint, and decreases total ownership costs.”(Davis, 2011). A further benefit of the common environment is that it reduces its staff’s workloads in terms of equipment, training, and logistics. Furthermore, it is estimated that this effort will help reduce \$4 million per ship and configuration in its current

implementation (Gavin, 2009).

Other programs such as the US Navy Virginia Class Submarine, US Army Victory Modular Standards, Modular Open Systems Architecture for Submarine Warfare Federated Tactical Systems, the Joint Surveillance and Target Attack Radar System, Common Ground Station program, Navy Area Theater Ballistic Missile Defense program, and the Open Mission Systems programs are discussed in detail in the remaining sections.

4.4 Hypothesis Overview – H3 & H4 – US and non-US Government department impacts

It is often not the speed of a discovery but rather the speed of the delivery. The current COVID-19 Crisis is an example of how vital it is for organizations, countries, and governments to rapidly evolve and deliver value and agility. The defense community has been under strong pressure to decrease costs, accelerate product development, and improve quality all while recognizing the need to achieve time and cost project targets. Recognizing the urge to achieve this, DoD led this change through MOSA. With hypotheses H3 and H4 we will dive deeper into how US government departments such as the Army, Navy, and Air Force have implemented MOSA and accordingly qualitatively observed positive improvements in working towards achieving these targets. Furthermore, we will take a step further and review global defense organizations who have used the MOSA success to implement similar strategies in their defense acquisition programs to also be able to have modular and adaptable defense systems.

4.4.1 US Army

The US Army often talks about Designing, Developing, Delivering, and Dominating the Army modernization strategy. MOSA is viewed as a warfighting imperative in the US Army. From acquiring new drones, helicopters, aircraft, and land defense systems to transmitting data seamlessly between all of these systems, MOSA is going to play a key role from being ahead in combat to long term upgrades. Through a modular and open systems approach focused on commonality and data sharing, the military will avoid expensive propriety solutions, greatly reduce costs, and decrease the time to field new competencies (Underwood, 2018). The Army is resolute on enforcing common standards, promoting open architectures, and developing a mindset change through MOSA. The military is advocating for modular and open systems all the way from the top (Serbu, 2019). On many occasions, the Army has emphasized that getting MOSA to work successfully is no longer a bureaucratic issue but a literal issue of life and death as the technology and directive that MOSA will implement will have a tactical, predictable, and strategic advantage over others.

Implementing MOSA in the Army has started aggressively. “The Army is determined to enforce common standards and common software in a way it never has before” (Freedberg,2020). From directing its stakeholders to implement MOSA through implementation policy, to identifying key technical enablers and program events as an implementation guide, the MOSA policy is in rapid utilization. The Army currently has multiple projects that support the MOSA initiative and a few programs such as Ground Common Infrastructure; Reference Architectures for Positioning, Navigation, & Timing; Open System Architectures for Future Vertical Lift; and the Improved Turbine Engine

Program have been recognized for their successful implementation within the Army network (Freedberg,2020). Figure 6 below gives an outline of the some of the successful programs that have implemented MOSA principles in the US Army (Alexander, Burnside, & Hurst, 2019).

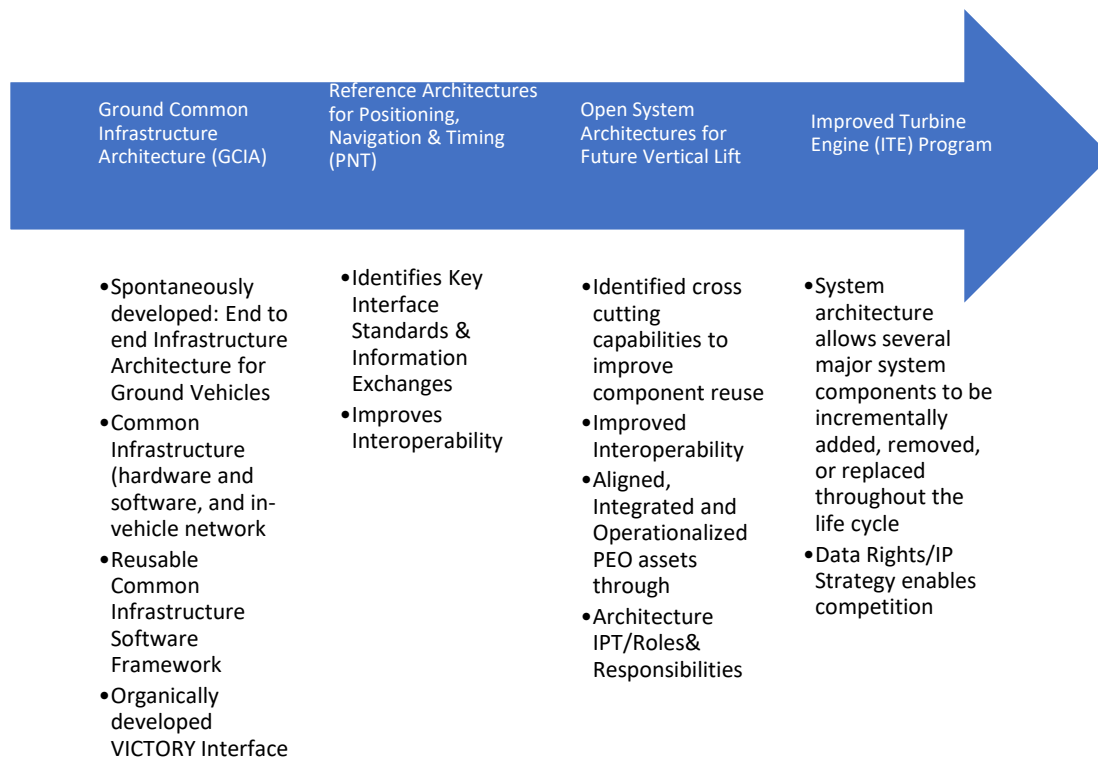


Figure 6: Successful Examples of MOSA within the US Army

As seen above, this is just the start of the work involved around MOSA and will only keep increasing over time. MOSA has led to increased ownership, agility, flexibility, and creating an exciting environment where the organization can quickly “test, fail, gain the learnings from that failure, and move those lessons forward” (Seawright, 2020).

There are a lot of reference standards and architectures that have led to the MOSA

implementation. Some of the fully mature standards and reference architectures within the Army are:

- Vehicular Integration for C4ISR/EW Interoperability
- Future Airborne Capability Environment
- C4ISR Open Suite of Standards
- Modular Active Protection Systems Framework
- Modular Open Radio Frequency Architecture
- Hardware Open Systems Technologies
- Integrated Sensor Architecture
- Robotics and Autonomous Systems, Ground
- Interoperability Profile

In conclusion, the Army continues to make tremendous strides with the MOSA strategy (Uppal 2020) and standards such as the Ordinance Interface Standards and Engagement Operations which are expected to reach maturity in the next few years and are expected to further help enhance the defense acquisition process. The US Army is pushing for modular and open standards through MOSA and requiring all programs and developmental activities for current and future development to include MOSA to the furthest degree possible (Uppal, 2020). MOSA also has helped the Army to move away from being tied to a specific vendor and integrating more plug and play solutions as well as making it simpler for small and medium scale businesses to work with the Army and help level the playing field (Keiser and Edwards, 2017). “The success can be seen in terms of increased warfighter effectiveness, enhanced levels of interoperability, and opportunities for cost savings simply by enforcing and employing a common set of

standards.” (Reese, 2015).

4.4.2 US Navy

In the U.S. Navy, the MOSA strategy has allowed its programs to accelerate deployment of updated technology within its fleet, focus on reduction of cost and cycle time, and promote a standard communication means among its defense systems and platforms in the Naval operational architecture. Two of the major successes with the Navy using the MOSA strategy have been the Littoral Combat Ship and the Virginia Class Submarine programs. Similar to the US Army, the Navy has promoted modularity and open architecture as a strategy for the future. The Navy however has a long history with the use of open architectures and modularity. The Navy had embraced open systems in small forms in a lot of its projects since 1994. Fast forward to 2015 through the Better Buying Power 3.0 mandate, modularity and open systems within the Navy have made significant inroads in implementing these principles. The MOSA mandate further helped strengthen and formalize that relationship.

Traces of open architecture date back to 2005 within the Navy. Certain projects that focused on open architecture and modularity such as the Marine Air Ground Task Force and the Consolidated Afloat Networks and Enterprise Services (CANES) helped reduce cost per ship by about \$2.9 Million and \$4 Million respectively (Minor, 2017). Below is a consolidation of the key MOSA efforts currently ongoing in the Navy.

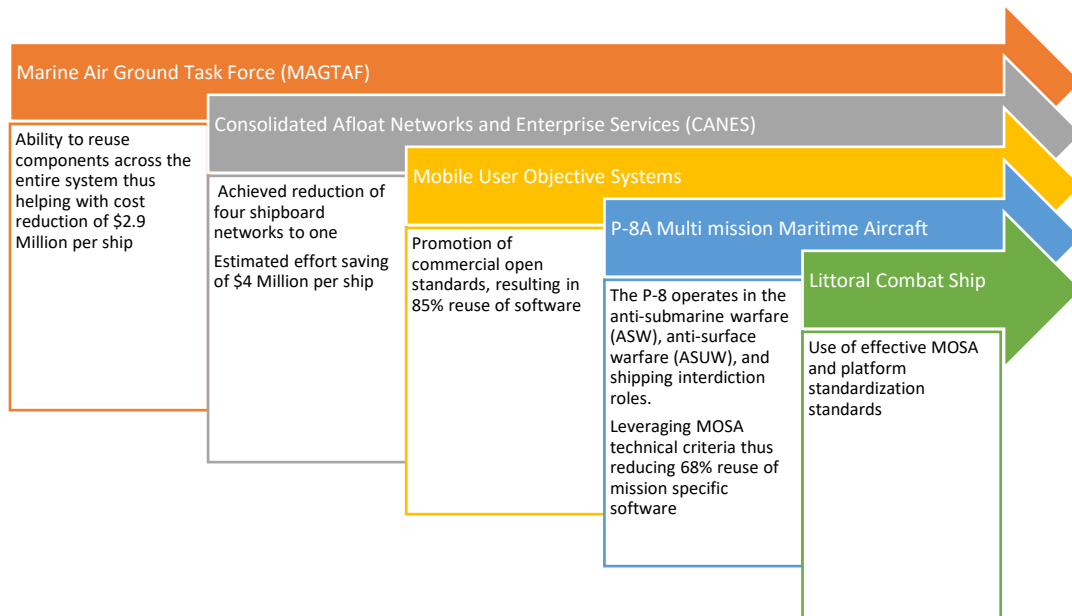


Figure 7: Successful Examples of MOSA within the Navy

A couple of other programs that have led the open system and modular efforts are:

- US Navy Virginia SSN 774 class which is a nuclear powered submarine that from the initial design has implemented an open system approach which has allowed the program to leverage COTS technologies, commercial standards, better integration of communication consoles, and increased use of existing open commercial standards (Strei, 2003).
- Acoustic Rapid COTS Insertion is a multi-phase program replacing legacy submarine sonar systems with a flexible, modular, and COTS based architecture that gives the US Navy a common sonar system across multiple submarine systems (Strei, 2003).

When one combines modularity with open-architecture approaches, the result is

an adaptable and flexible warfighting machine which is able to support any threats or technological refresh opportunities. From the above discussion similar to the Army, the Navy has devoted a great deal of effort in its warfighting ability through the MOSA architecture and as and when more of these successes come to life, there will be a push to evaluate each more specifically in terms of Return-on-Investment metrics.

4.4.3 Avionics within the DoD

The literature review discovered several avionics systems that were reporting success with modularity. “MOSA marries the business and acquisition process of the open systems with the technical tools and processes to enable use of modular designs to achieve the MOSA benefits” (OUSD, 2017). With the increased collective efforts between industry leaders, the defense administration, and the defense community, it is vital to understand the beneficial characteristics of these successes. There has been tremendous success from initiatives such as the Unmanned Systems Control Segment Architecture, Open Mission Systems Initiative, and the Future Airborne Capability Environment approaches (Tokar, 2017).

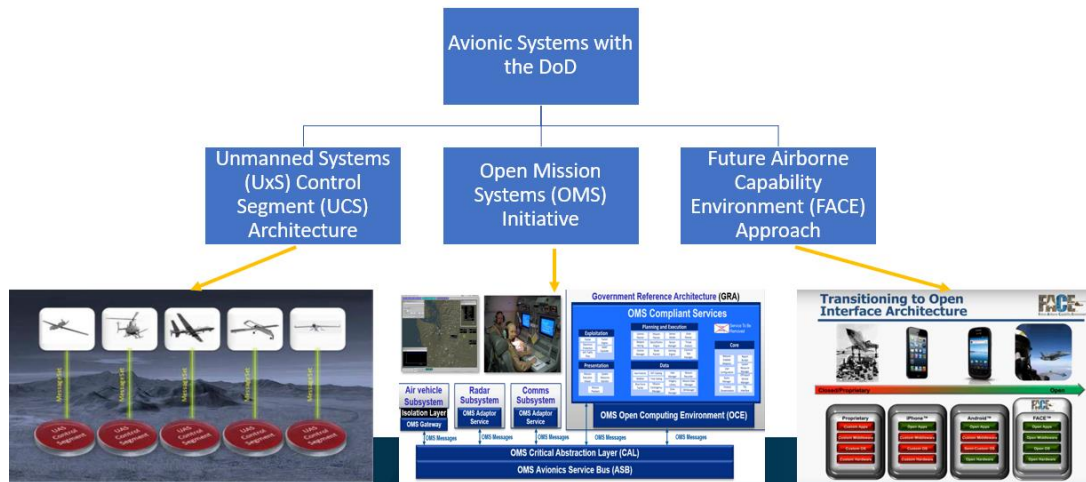


Figure 8: Overview of the Avionic Strategies

While the US Army and Navy have made amazing strides with the MOSA strategy, a large part of the successes do come out from the aviation mission which includes systems such as drones, helicopters, and other aerial combat systems. The next few sections will dive further into each application to better understand what they are about and the successes coming out of applying the modular strategy.

1. Unmanned Systems Control Segment (UCS) Architecture

The Unmanned Systems Control Segment Architecture was established by the UCS technical governances team to focus on Unmanned Aircraft Systems. The endeavors were later shifted to the Society of Automotive Engineers Unmanned Systems Committee in the first quarter of 2015. With the increasing use of unmanned technology, the UCS architecture developed as the DoD's information architecture for managing robotics and autonomous systems for systems in the land, air and sea (Ernst 2016). Thus, the nomenclature was modified to be called the Unmanned Systems Control Segment Architecture.

As with other existing defense systems, prior Unmanned Aerial Vehicles systems were manufactured and obtained using traditional proprietary approaches with the goal to break away from this traditional approach and move towards an open-source effort for the applications. The end goal was to drive down life cycle costs and promote interoperability, similar to the lines of MOSA.

The UCS's first objective was to standardize a common message protocol to communicate with unmanned air systems using a standardized command and control center. Figure 9 (Tokar, 2017) showcases the open system interaction model across various defense applications and services.

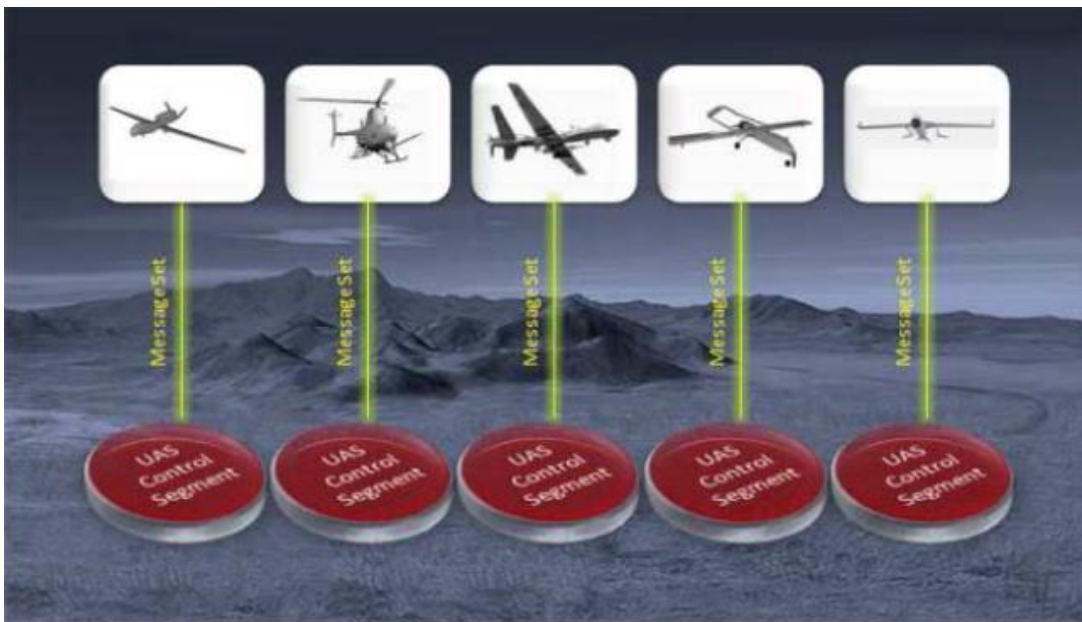


Figure 9: UCS Architecture

The UCS focuses on three major facets with respect to interoperability and to focus on the relationship between elements - conceptual interoperability, to better address functionality – pragmatic interoperability, and to tackle encoding of data – semantic

interoperability. Unmanned Aerial Systems have become a pivotal focus of continuous improvement in the aviation segment. "Unmanned Aerial technologies have incredible promise and potential to not only provide great economic benefit for the American people, but also to enhance safety and security for our nation. We need a strong, secure domestic UAS manufacturing base to ensure American leadership in this critical field", Acting Undersecretary of Defense for Research and Engineering Michael Kratsios (Vergun, 2020).

With heavy investment exceeding ten billion dollars annually, there is a strong focus on developing future capabilities of a more robust architecture. "The Navy used an open systems approach at the start of development for the air vehicle, ground control station, and payloads (i.e., cameras and radar sensors) for three of its four current and planned UAS and anticipates significant efficiencies" (GAO, 2013). The US Navy was the forefront adopter of open systems approach for the unmanned system programs. The Small Tactical Unmanned Aircraft System, the Triton, and the Unmanned Carrier Launched Airborne Surveillance and Strike programs adopted open system approaches from the start of their schedule and all the programs were able to achieve cost and schedule benefits. Whereas the Fire Scout, the Gray Eagle, the Hunter, the Shadow, the Reaper, Global Hawk, and the Predator failed to adopt open systems during the early stages and were not able to leverage benefits such as upgrades and integration of COTS. (GAO, 2013). The UCS architecture was based on Open Business Model principles and is being performed in conjunction with the Better Buying Power 3.0 initiative.

The UCS architecture will help to drive reduction in dependence on single source vendors, promote component reuse, enable cost savings, and reduce integration time

(Gregory, Batavia, Coats, Allport, Jennings, Ernst, 2013). An added benefit is that the UCS architecture allows for small business, academia, and similar markets to access and promote this technology. Currently, the UCS architecture is being adopted into several DoD programs. The UCS architecture will help create a varied and open ecosystem for modular and open applications within the DoD. Furthermore, this will support the MOSA initiative with targeting cost, focusing on affordability, and promoting a repository of great open system learnings within its applications.

These include, in the Navy, Littoral Combat Ships, Common Control Station, Multi-Operator Control Unit, Fire Scout Tactical Control Station, and the Unmanned Surface Vehicle programs of record. In the Army, they include that Joint Multi Role Rotorcraft, Universal Ground Control Station, and the Bi-directional Remote Video Terminal, and in the Air Force, the ADM Global Hawk GCS Upgrade, and the Block 50 Ground Control Station projects (Ernst, 2016).

The conclusion that can be drawn from the UCS initiative is that there is an uphill task to develop specific guidelines and standards that will support interoperability of systems which will give the ability for military equipment or groups to operate in concurrence with each other. These findings clearly show that some of MOSA's ideas are taking hold in the aviation realm and are here to stay. The development of these standards has not yet however allowed for increased interchangeability by different military groups or for them to work together and that is where there is a need for further development.

2. Open Mission Systems (OMS) Initiative

The goal for the Open Mission Systems initiative is to create a non-proprietary

open architecture to enable affordability and to drive a defense acquisition model that fits the new world of acquiring defense systems. The OMS is one of the initiatives that goes hand in hand with the MOSA principles. The benefits OMS is trying to achieve are very similar to what MOSA intends to accomplish. “Adopting open architectures in systems that the Defense Department buys from industry can reduce costs and facilitate competition.” The integration of the MOSA strategy is to ensure that there is a common starting point for various departments to leverage open and modular systems effectively as well as share common principles and successes (Bratton, 2019).

With the directive of the Better Buying Power 3.0 and other governmental guidelines, OMS will help reduce development and life cycle cost while providing a parallel benefit of easy upgrading and expanding current defense capabilities. As per (Tokar 2017), the OMS architecture will deliver benefits such as increased interoperability, affordable technical refresh, streamlined integration and an opportunity for service reuse. “Almost 75 percent of Defense Department acquisition strategies implement open systems architecture across all services and agencies.” (Pellerin, 2014).

The Open Mission Systems standard (OMS 2015) was developed by the US Air Force, with the intention of focusing on service architecture concepts. The main focus of their work was to provide an ability for systems to interface with subsystems through the use of an Avionics Service Bus. The OMS specification helps with the better understanding of specific requirements and developing robust compliance strategies. “The Air Force is expanding its Open Mission Systems standard because it is working so well, Lt. Gen. Robert Otto, deputy chief of staff for intelligence, surveillance and reconnaissance” (Whittle, 2015). The Air Force’s goal is integrating the competencies of

the OMS standard to speed up the advancement in aviation systems.

An overview of the OMS architecture is provided below in Figure 10 which highlights the key modules and boundaries (Falcone, 2018). As we can observe below, the architecture supports multiple services as well as the Avionics Service Bus protocols and assists with migration and advancement of compliance in open mission systems. The reference architecture further emphasizes and identifies compliance specific methodology required to support compliance for other services.

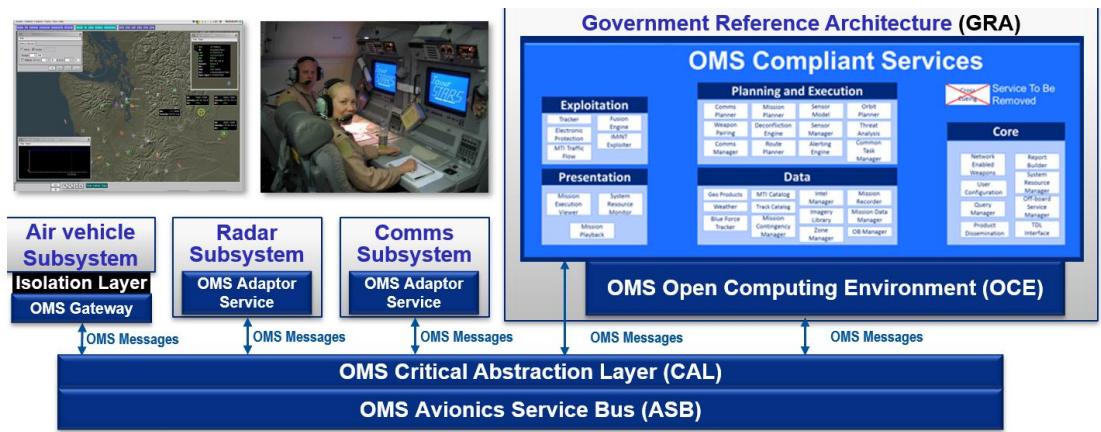


Figure 10: OMS Architecture Reference

The MOSA standard helps complement the OMS architecture by providing additional requirements and guidance to support the usage of OMS in defense systems. “Open avionics systems are crucial to empowering the cost effective, and timely introduction of new air power capabilities in platforms that will serve for decades. Service oriented concepts combined with embedded open system methods will deliver future open avionics technologies and architectures” (Tokar, 2017).

The US Air Force, Navy, and Space programs have adopted and started utilizing the open mission system architecture across their respective projects as well. Some recent

examples of successful implementation are in the integration of a new Automatic Target Recognition service into an already OMS-compliant weapon system and the sharing of Common Operating Picture data between an air operations center and a space operations center. Continuous collaboration with the defense community and industry partners will help to develop further understanding of the capabilities and limitations of open systems.

The outdated strategy of relying on single source suppliers to manufacture huge quantities of defense systems and be dependent on their post-production services will no longer work (Wilson, 2015). Organizations have become smarter and demand they own the Intellectual Property and have the ability to integrate upgrades as they see fit has become the industry norm.

OMS is not a strategy to replace systems engineering efforts but instead meant to heighten system engineering methods by deploying OMS strategies to systems, subsystems, and services. MOSA is a larger part of the open system strategy that incorporates open system initiatives such as the Open Mission System, Unmanned Common Architecture etc. The OMS tactic helps by providing a starting place to address some of the challenges in the avionics segment. The goal remains very synced to what the government wants to achieve – an open architectural standard promoting streamlined mission integration, empowering affordability, and enabling service reuse and interoperability, all the while collectively working industry leaders to deliver this benefit to defense systems. One of the key areas that remains a gap in this approach, however, is ensuring cyber security while promoting open standards. It is one of the tougher problems to solve where there is cooperation and mutual consent required between industry and government leaders and leveraging industry best practices into these efforts. As open

systems are encouraged, there is an increased risk to cybersecurity. In fact, the US government estimates that nearly \$600 Billion which equates to roughly 1% of the global gross domestic product is affected and lost through cyber threats and theft (Lopez, 2020).

3. Future Airborne Capability Environment (FACE) Approach

The FACE approach was developed in 2010 to focus on developing an open aviation strategy for military avionics platform types. A decade later, FACE is a real time software group that includes defense personnel, industry suppliers, and academia all working together to build an affordable software system that drives the rapid integration of COTS and promotes innovation. The goal of FACE is to standardize open system approaches in avionic systems and being in common solutions. The FACE approach allows an opportunity for the US government and its vendors to work together and develop open system solutions and best practices specific for aviation systems.

“The US DoD continues to leverage open architecture solutions that help improve speed, flexibility, and costs. The FACE approach is one of those methods through which open system standards are leveraged and avionics hardware is integrated to ensure operations are more secure, portable, and strong” (Seawright, 2020). The FACE technical standard was built on the motivation of the advances in mobile devices using common environments. “With the creation of the FACE Technical Standard, the DoD and our industry partners in the FACE Consortium are setting the direction for software portability and interoperability. This standard will enable reduced costs and speed delivery of technical innovation to the Warfighting community.” (Tescher 2012).

As per Dave Nieuwsma, Vice President and General Manager of the Airborne

Solutions Division of Collins Aerospace, “Modular open systems are very important to the industry, our military customers, and the warfighter”. “We have seen the benefits of such systems within military rotary wing and tanker / transport aircraft and are pleased to participate as a sponsor member of the FACE Consortium. This, along with the forthcoming conformance certification process, will promote the use of widely adopted open industry standards, minimizing the cost and schedule of technology insertions and capability upgrades.” (Tescher 2012).

“I have been a supporter of the FACE Consortium and its efforts since I was first briefed on their activities,” said Colonel Anthony W. Potts, Program Manager, PM Aviation Systems, U.S. Army. “Software portability is essential to reducing costs and expediting the fielding of new capabilities to our aviation warfighters. I have directed the FACE standards be applied to the design and development of the Improved Data Modem Open System Architecture (IDM OSA), as the Army’s first application of this important standard.” (Tescher 2012).



Figure 11: Synopsis of the FACE Approach

The objective of the FACE Consortium is to standardize methodologies for applying open standards in aviation systems, decrease operational costs, enable a robust architecture, increase competition within vendors, and drive innovation – all similar to what MOSA wants to achieve. The development of the FACE standard will deliver a standardized framework that will allow the military to transform its existing architecture to a universal and common operating environment to support aviation missions within the DoD. An overview of the FACE approach is shared through Figure 11 (FACE, 2018).

The FACE Reference Architecture is comprised of five segments:

- Operating System Segment that focuses on the foundation and the services provided
- I/O Services Segment which focuses on the interface set

- Platform-Specific Services Segment which help provide health and data services and also assist by allowing graphics to interface
- Transport Services Segment which leads communication protocols
- Portable Components Segment where all the components exist

“The open system architecture helps by decreases integration costs that are usually present when adding new enhancements. A standardized and consistent use of common open and modular standards across aviation defense systems will help to eliminate further reliance on exclusive solutions. The relationship can be viewed through Figure 12” (Tokar, 2017).



Figure 12: FACE Integration Approach

The U.S. Air Force, Army, Navy, Special Forces, and coalition forces support the FACE standard through over 20 procurements with FACE requirements. Along with the

FACE standard, there is a consortium that includes about 70 organizations and 900 individual contributors supporting a growing relationship between vendors and the government to develop open and secure architectures. Organizations such as Boeing, BAE Systems, Bell Helicopter, and General Electric represent some of the Tier I suppliers and key members of the group. "FACE provided an opportunity to deploy products that helped decrease development cycles and integration issues" (Brabson and Anderson, 2015). Applying the FACE methodology will help by improving limited resources, increasing new commercial opportunities, and supporting the United States and its partners in sustaining a strategic military advantage (Downing, 2017). Overall, the FACE initiative works to deliver similar objectives to the MOSA strategy, and the end goal remains to incorporate open systems into existing and future defense systems with ease to deliver agile, flexible, and modular defense systems.

4.4.4 Global Overview

This praxis' focus on other country's adaptation to modularity is essential to its research methodology as it gives a strong understanding that modularity and open systems are becoming the new norm in all new product development projects regardless of commercial or defense interest. "In general, MOSA initiatives are adopted for three primary reasons: (1) to avoid being left behind as others find new opportunities, (2) to take advantage of new methods to improve internal corporate efficiency, as well as to (3) increase market share and increase profits" (Guertin, 2018).

The United States being the forefront in innovation and the application of the MOSA strategy has inspired other governments and defense organizations to adopt modular and open strategies for their defense systems as they also face similar issues.

Some of the early adopters of the MOSA strategy have been the UK MOD, Australian DoD, and the CAF.

1. MOSA & the Canadian Procurement Strategy

The strides and developments made by the US DoD have led to studies and similar strategies applied globally (Christianson, 2018). The CAF and government have had similar problems with their defense systems procurement programs failing to meet schedule and cost targets (Stone, 2015). The significance of getting defense procurement strategies right cannot be understated and the lack of ability to deliver and maintain defense systems possess a great threat and risk not only to the government and defense organizations but also compromises the defense capabilities of the country. While smaller Canadian programs have been acquired in more timely fashion, a lot of the larger acquisitions have faced major delays and cost overruns, similar to the US DoD. These challenges have led to political, media, governmental, and public concerns with respect to the departments being able to appropriately manage Canadian taxpayers funds (Auger, 2016). While it is acknowledged that flaws within Canada's defense procurement process leading to project delays and cost overruns are a "perennial burden on both government and industry," (Stone 2015), the Praxis will highlight the effects and evolving challenges associated with global defense procurement strategies.

As costs associated with defense systems keep rising, the MOSA philosophy focuses on leveraging IT trends and open system architectures to improve innovation and incorporate new technologies all the while reducing costs. The Canadian government made some aggressive moves to improve their defense acquisition strategies through launching a centralized shipbuilding strategy in 2010 and a defense procurement strategy

to increase efficiency and accountability (Auger, 2016).

The CAF has acknowledged that modular and open strategies show promise of reducing life cycle costs and incorporate reduced risk while enhancing flexibility and adaptability while procuring defense systems (Christianson, 2018). Many Canadian defense system platforms such as the Sea King, Aurora, the CF-18 Hornet, and Halifax class frigates are all in service after three decades and well beyond their intended lifespans. Such a trend is further exacerbated by the fact that the procurement process itself can add an additional 10 to 15 years between identification of requirements and delivery; meaning that a defined requirement from identification could be expected to remain relevant for as much as half a century if no planned upgrade is incorporated (Williams 2006). Since defense systems come with a significant investment, it is understandable that many of these systems are active and over their intended lifespan. However, with the pace at which technology is evolving, it is essential that defense systems of today are more agile and flexible and that they support the diverse missions, threats, operating environments, and changing governmental policies. Recent history has witnessed a near exponential increase in computing and networking capabilities, a trend which is fully anticipated to continue as suggested by concepts such as Moore's Law and Butter's Law which argue computing and communications capacity double every 24 and 9 months respectively (Simonite, 2020). Moore's Law anticipated half a decade ago that technology in transistors per chip would double every two years and served as a rule of innovation as products and services advanced. Fifty years later, the law still holds good. Butter's Law predicted the amount of data in a fiber cable would double every nine months and this led to work in optical fiber cables. Analysis across multiple weapon

classes has demonstrated that per unit costs have increased significantly faster than the rate of general inflation or contemporary Gross Domestic Products, in many cases at rates exceeding 5-10 percent per year or higher for technologically intensive capabilities (Kirkpatrick 2004). This has led to a decrease in purchasing power for defense programs around the world and has also led to a focus on platforms being more adaptable rather than frequently replacing defense systems platforms as per traditional approaches.

2. U.K. Ministry of Defense's Approach to Modularity

The United States and Canada are not the only nations to pursue modular and open architecture initiatives. Observing the early signs of progress of the US DoD such as Victory (US Army), the Future Airborne Capability Environment (US Navy), and the Open System Approach, the UK Ministry of Defence has been prioritizing similar approaches with the Generic Vehicle Architecture (GVA) program.

The United States has always been the vanguard when it comes to defense and defense spending. Understanding the success and struggles of other countries who have also adopted open systems and modularity is essential as it gives a further understanding if defense organizations are able to better measure and justify a modular strategy. With heightened implementation of MOSA there will be better data to quantify how organizations are able to better qualify defense systems and whether success lies in systems and platforms, sub systems, or is just constrained to off the shelf products. There lies an opportunity to measure defense programs and compare their project performance success to defense systems across the globe. This comparison will further help reiterate if modularity and open systems are actually delivering the benefits that they had intended

to. The benefits of MOSA are clear. It helps with easier integration, enhanced flexibility, faster upgrades, and great long-term value (Clark, 2019).

The UK MOD has made great strides in multiple projects since April 2018 in including modular and system approaches within their acquisition projects. The introduction of the mandatory Open Architecture Key User Requirement, developing of the new open architecture vehicles, and the Land Open System Architecture initiative have all been positive steps towards a more agile defense strategy (Clark, 2019). UK has further articulated such objectives within its System of Systems Open System Strategy and has begun implementation of MOSA as part of the both the New Generation Submarine Command System and the General Vehicle Architecture for Land Vehicles (Secretary of State for Defence, 2012). The latest document, “National Security Through Technology: Technology, Equipment, and Support for UK Defence and Security” has re-accentuated that the UK's defense and security requirements must be acquired through open competition from the domestic and global marketplace, and through buying commercial products such as COTS which use open standards (Ministry of Defence, 2012). The MOD has also published similar standards to MOSA called Systems of Systems Approach (SOSA), that largely focuses on acquiring defense capabilities through the use of modular and open systems. “SOSA offers nine design principles for decision makers to guide behaviors throughout the acquisition lifecycle and all nine have the potential to be met through the adoption of an open approach” (Henshaw 2019). The GVA was built with the Royal Army’s concern that the nature of the threat is constantly evolving and by the time the army has adapted highly technological capabilities into their vehicles, the enemy has already countered and advanced their technology. Thus, the

MOD focused on a more open and modular response to better address the system upgrades and integration challenges.

Using an open system approach and common interface, the MOD was able to successfully integrate a common functionality to increase land vehicle effectiveness thus enhancing adaptability and introducing plug and play into their land defense systems.

“The MoD has set a world-leading example of how to manage defense procurements in the face of new threats as the enemy becomes increasingly shrewd in leveraging their asymmetric capabilities. With GVA and Def Stan 23-09, the MoD has a procurement strategy for delivering military vehicles that are coherently designed, while maximizing utility and enabling field command to get the best logistically out of its assets. The architecture empowers field command to plug-and-play so as to adjust to rapidly changing threat profiles.” (Hunt 2011).

From the above discussion, it is evident that modularity and open systems are playing a big part in shaping the procurement strategies for organizations and countries outside the U.S. and that like the US, other governments have mandated modular approaches. The research did not reveal, however, how these other governments were measuring the program performance effects of modularity only that the effects were positively felt.

4.5 Hypothesis Overview – H5 & H6 – MOSA’s impact on the commercial sector

The reason the praxis is including MOSA’s effects on commercial vendors adopting MOSA standards is because during the literature review, it was evident that the open systems mandate by the US Government will require a new way of working with its

suppliers. The US DoD has been too reliant on single source vendors. “A recent GAO report revealed that the average acquisition for a major Defense weapons system experienced a 38% cost growth from original estimates and a 27-month schedule overrun” (GAO, 2012). Vendor lock often leads to proprietary data and intellectual property rights controlled by the supplier and any changes or upgrades came at a higher cost due to the ownership. With the execution of MOSA, the government has made it a fair playing field, controls the rights, and prevents misuse from a single vendor. One of the ways MOSA is trying to better adapt to this is by creating increased competition within suppliers and vendors of the defense systems. The DoD is demanding that its suppliers adopt to a more flexible and open platform so that its defense systems can be used for multiple missions needs instead of highly concentrated needs. The MOSA strategy has thus led to the US DoD ending reliance on single source vendors and working towards increasing competition within its ecosystem. “A recent study by the Center for Strategic and International Studies estimates that from 2001 to 2015, 17,000 companies ceased to be prime vendors for DoD.” (McCormick, 2018). This is a positive effect that is allowing the DoD to increase competition amongst its vendors and thus driving better affordability within its projects.

The MOSA methodology warrants a better understanding of how the DoD is focusing on transforming its acquisition process to be more adaptable. With the increased spending over the years on defense systems and the continuous trend of defense systems being over budget or over schedule, there was a strong review of programs by the DoD and the Government Accountability Office to ensure that these problems will not persist.

The HForce by Airbus is one of the latest generation air weapon systems that meets

the requirements of a defense system of the future. The HForce is a modular and flexible helicopter that has been developed with off the shelf solution needs to cater to multiple mission needs. From intelligence to surveillance to combat recovery, defense agencies can gain flexibility in their use of systems like these as well as transition to another mission need quickly. “Airbus Helicopters recently changed the situation of single purpose helicopters to releasing helicopters with multi-purpose mission computer platform such as HForce that can be fitted into any military version of Airbus’s commercial helicopters, including the H125M, H145M, and H225M helicopters, converting any commercial helicopter into an armed, battle-ready unit for military operations. Designed as a modular and incremental system, HForce offers a large choice of weapons (air-to-air, air-to-ground, ballistic or guided) to support any operational need.” (Downing 2018). Similar to the HForce defense system, the trend of suppliers delivering complex, modern, and modular systems are here to stay and will support the vision of defense organizations.

4.6 Summary of Results

The chosen praxis research methodology was insightful in leading towards broader trends coupled with examples on modularity and open systems. While the first quantitative phase of the research failed to develop statistically significant proof of this effect, proof was established qualitatively by pivoting towards assessing known successes and providing key insights on where and how measurement of MOSA’s effects could be improved. Successful examples have paved the way for more confidence in the DoD community toward the adoption of MOSA. Avionics systems have had a great deal of

success in the realm of open modular system implementation. This further led to the discussion on whether land and sea defense systems had seen similar successes. One of the key findings of this research, however, is that while qualitative anecdotal evidence of modularity's success abounds, without quantitative measures specific to MOSA's effects, it remains incredibly hard to compare program performance success and attribute it to the MOSA strategy. Without good measures, justifying the investment in modularity and assessing when and where it is most effective can also be a challenge. "Open Systems, modularity and use of Commercial Off The Shelf products are good systems engineering principles. Just as it is impractical to separate the cost savings attributable to OSA from those attributable to COTS, it is equally impossible to separate cost savings attributable to good systems engineering and effective program management." (Worthington, 2012).

Table 5: Summary of Hypotheses

Hypothesis	Summary of Hypotheses	Supporting Examples and summary
H1: If MOSA is successful, then the expectation is to see good cost and schedule effects available from program performance stats attributable to modularity.	Not Supported.	Reviewed 86 Major Defense Acquisition Programs in 2017. Overall, the deep dive showed that currently it is impossible with the published MDAP performance data alone to isolate the effect of MOSA on defense acquisition programs and measure its impact. Since defense acquisition programs are multiyear and there are so many confounding variables associated with the scale of such projects, there just was not sufficient data to show that MOSA specifically has led to the success or failure of a specific programs
H2: If MOSA is successful, then there is evidence that government programs will claim so and not claim otherwise.	Supported.	Relevant Programs - Global Hawk and MQ-9 Reaper unmanned vehicles; AH-64 E Apache, and HH-60W Black Hawk Rescue Helicopters; U-2 Dragon Lady; F-22; AV- 8B Harrier jets, B-52 Stratofortress and the B-58 Hustler. Additional insights - the Consolidated Afloat Network Enterprise Service, US Navy Virginia Class Submarine, U.S. Army Victory Modular Standards, Modular Open Systems Architecture for Submarine Warfare Federated Tactical Systems, Common Ground Station program, Navy Area Theater Ballistic Missile Defense program and the Blue Guardian and open mission systems program.
H3: If MOSA is successful, then there is evidence that other US and non-US government departments will also require it.	Supported.	Army
H4: If MOSA is successful, then other US departments and	Supported.	Programs - Ground Common Infrastructure, Reference Architectures for Positioning, Navigation & Timing, Open

non-US government departments will show evidence of its success and not claim otherwise.		System Architectures for Future Vertical Lift, and Improved Turbine Engine Program - Standards - Vehicular Integration for C4ISR/EW Interoperability, Future Airborne Capability Environment, C4ISR Open Suite of Standards, Modular Active Protection Systems Framework, Modular Open Radio Frequency Architecture, Hardware Open Systems Technologies, Integrated Sensor Architecture, Robotics and Autonomous Systems, Ground, Interoperability Profile Navy - Programs - Littoral Combat Ship, Virginia Class Submarine, Marine Air Ground Task Force and the Consolidated Afloat Networks and Enterprise Services, Mobile User Objective Systems, P-8A Multi mission Maritime Aircraft Air Force & Aviation – - Initiatives - The Unmanned Systems Command and Control Architecture, Open Mission Systems Initiative, Future Airborne Capability Environment Programs – - The Unmanned Systems Command and Control Architecture - In the Navy, Littoral Combat Ships, Common Control Station, Multi-Operator Control Unit, Fire Scout Tactical Control Station and Unmanned Surface Vehicle PoR. In the Army,
---	--	--

		Joint Multi Role Rotorcraft. Universal Ground Control Station and Bi-directional Remote Video Terminal. And in the Air Force, ADM Global Hawk GCS Upgrade and Block 50 Ground Control Station projects. • Open Mission Systems Initiative - Some recent examples of successful implementation are in the integration of a new Automatic Target Recognition service into an already OMS-compliant weapon system and sharing of Common Operating Picture data between an air operations floor and a space operations floor • Future Airborne Capability Environment Consortium and its comprised group of 70 major corporations and 900 individual contributors. Some major members include, Boeing, Bell Helicopter, Wind River and UTS Areospace Systems. • Global Overview – Early adopters of MOSA are the UK Ministry of Defence, Australian DoD, and the Canadian Armed Forces • Programs – The Canadian acquisition of CF-18 Hornet Aircrafts, MOSA strategy led to the implementation of the Generic Vehicle Architecture program in the U.K. Ministry of Defense.
H5: If MOSA successful, then there is evidence that the commercial sector has/is adopting it.	Supported.	HForce by Airbus
H6: If MOSA successful then there is evidence that the	Supported.	

commercial sector is having success with it and not claim otherwise.		• H125M, H145M, and H225M helicopters, converting any commercial platform into an armed, battle-ready unit for military operations • HForce offers a large choice of weapons (air-to-air, air-to-ground, ballistic or guided) to support any operational need
---	--	--

Chapter 5 – Conclusions and Recommendations

5.1 Introduction

Chapter 5 will summarize the praxis and present how this research contributes to systems acquisition. This chapter also provides recommendations for future research opportunities.

5.2 MOSA Discussion

From the overall praxis a better understanding on the reason behind the mandate on the MOSA strategy and what it is all about can be gained. Open systems and modularity carry a large interest due to the potential of lower program costs, increased integration of COTS, and ease of integration. While the government has taken multiple steps to address the failure to meet cost and schedule targets, MOSA brings a promising turn on assisting with defense acquisition. Since the Better Buying Power 3.0 mandate in 2015, not a single study has looked at the overall MOSA performance from a Return of Investment perspective. Sooner or later, the DoD will want to understand how the MOSA strategy is performing and whether it is achieving its intended goals. From a project level, MOSA has successfully increased industry collaboration, developed consortiums that included the best minds and skills globally working towards a goal of increased modular and open system integration, and increased buy in on all the aspects of the architecture (Worthington, 2012).

Some issues the DoD might run into while trying to evaluate this would be the similar problem this praxis intended to achieve initially which was through a quantitative multi-year analysis and whether programs that use the MOSA strategy are performing better in terms of achieving cost and schedule targets versus programs that currently do not

employ the use of MOSA. At what level and detail will the evaluation take place, platform, system, sub-system, service, COTS integration? Whether a distinction will occur between the Army, Navy, and the Air Force or will all the programs be combined under the DoD umbrella?

With all that said, this praxis contributes a synoptic view of MOSA not only within the DoD but also its impact on a global foreign military defense and commercial organizations standpoint. Thus, this work is a unique and valuable contribution. This praxis compiled a lot of various examples and summarizes that MOSA as a strategy has had a lot of great impact to project performance from various branches within the DoD. Along with examples, there are a lot of positive case studies with cost and time savings in projects, but they are still not clearly available or stored in a separate repository for better access. An opportunity exists to share proven practices and lessons learned so different teams do not repeat similar mistakes and help the organization succeed as one.

From an overall praxis view, the research objectives and the problem statements have been addressed through the different discussions and analysis. As per the problem statement, *“An organized search for the representation of evidence of the success of the MOSA strategy is necessary to sustain and characterize MOSA and what about it is working . Such an effort is also needed to develop recommendations that will enable its success.”* The praxis was able to demonstrate multiple views of MOSA. Modularity, open systems, and defense acquisition processes are all described through the literature search and the body of knowledge. The praxis was able to achieve its objective of introducing the concept of MOSA to its readers, analyze the impact of MOSA across the different departments in the US DoD as well as other sectors through the use of examples and

literature compilation, and finally demonstrate the emergence of MOSA as a strategy that will transform the acquisition of defense systems.

From the research and literature review, there is clearly a limitation on the understanding of modularity and openness in the acquisition process. The DoD scope of MOSA methodology needs to include not only program managers but also a larger ecosystem to appropriately adopt MOSA strategies. “MOSA must become commonplace, with design patterns, instantiations, tool use, measurement, etc. Elements of MOSA practice are found but are not habitually documented” (Zimmerman, Ofori, Barrett, Soler, & Harriman, 2018). Obviously, there is a need to include long term goals and strategies specific to MOSA and its principles. From this, there is a need to specify, educate, and understand the implementation of MOSA currently and to integrate future adoption with MOSA standards, principles, and a more robust chain of communication. In order to gain the maximum advantage from modular systems, organizations will need to focus on how modularity can work best for their needs and requirements. (Pakkanen, Juuti, & Lehtonen, 2018). The research also revealed a strong requirement to include a phase to evaluate modular strategies in the initial planning of the life cycle process along with formalized pass-fail criteria so that vendors and the armed forces are on the same page when it comes to defense acquisition. The research suggests that another point is to implement MOSA specific training. While it is good to have documentation as a reference point, it is essential that all the stakeholders get additional training specific to MOSA to ensure compliance. Incentives, both positive and negative, should be considered to help overcome resistance to change and the perceived risks of implementing a new approach. What to measure, how to measure, and with what

measures still remain key challenges. Lastly, the research exposed the need for a formal repository showcasing effective knowledge management, case studies, and lessons learned from all legacy and current programs that use modularity. One of the best ways to describe concepts such as modularity and flexibility is similar to how quality was described a few decades ago, “vague and difficult to improve yet critical to competitiveness” (Upton, 1995).

5.3 Other Observations

Open systems and modularity bring a large appeal due to the potential of lower program costs, increased integration of COTS, and ease of integration. However, with greater connectivity there will be a higher risk of vulnerability due to the openness of a system. None of the literature surveyed mentioned this key detail. Clearly, there is a need for more research into how open systems and modularity can not only facilitate more secure defense systems but also help defense programs facilitate technology refresh, increase innovation, and enable cost savings.

To ensure that US DoD keeps its competitive advantage, researchers and strategists have followed leading trends in technology such as lean and agile methods, smart manufacturing, concurrent engineering, and autonomy. These approaches have failed to address the root cause of why the US has fallen behind in fielding new technology efficiently and affordably as compared to its competitors. The MOSA initiative helps solve this problem by requiring a fresh acquisition model focusing on flexible system architectures that allow quick integration of new technologies and to continually upgrade systems allowing the US to adapt to new threats under the ever-increasing economic pressure. The main goal of MOSA is to increase affordability and improve the

timelines of defense acquisition programs so that they can be mission ready.

While it is premature to decisively say that MOSA as a strategy is successful in achieving all its intended benefits such as enhanced competition with vendors, reduction in cycle time and costs, ease of integration, and other similar benefits, early indications suggest that the move to modular and open systems through MOSA is on the right path. That being said, there is a strong need to expand the use of MOSA and modularity across systems, sub-systems, interfaces, and platform architectures. It is critical from the above work that there is a need for more development of modular standards and assessments. There is substantial prospect for further assistance on modular integrations and analysis that will help transition the use of modularity into other branches of the government and other industries.

5.4 Limitations

Modularity, open architecture, and MOSA are all very broad terms and can lead to various interpretations. Further, there may be programs within the DoD that have already applied some level of modularity or are currently applying modular approaches using different terminology. The US DoD is the largest defense engineering organization in the world. Due to this, there are a lot of great processes, expertise, and high-level system engineers that practice to create large scale, interdependent, and highly complex defense systems. MOSA is one behavior that exists to assist with building successful highly technological defense systems while trying to balance schedule and cost goals. Due to the limitations placed on US DoD programmatic information available to private citizens, not all the information needed to study MOSA's impact on programs is readily available.

This research also does not indicate that MOSA is the only factor that could affect program acquisition. While there may be multiple confounding variables impacting a defense system's acquisition and its acquisition cycle can be multiyear (and usually consist of multiple sub-systems), the goal of this research was to provide insight on MOSA, the lack of literature and scholarly articles on this topic, and to begin to close that gap. With respect to quantifying if open systems can prove cost savings quantitatively, this remains a challenge since there are many factors in acquisition programs in parallel, data for potential cost saving for alternative procurement approaches does not exist and cost savings are primarily just estimated for a business justification (Worthington, 2012). The goal was also to shed light on how the measurement of MOSA's effects can be done better so that the positive influence of MOSA can be effectively assessed and promoted.

5.5 Contributions to the Body of Knowledge

Currently there is very limited research pertaining to how effective the MOSA strategy has been in the DoD acquisition process. Until now there has not been any significant work summarizing the efforts of MOSA from a holistic view. From the research and literature review, there is a limitation on the understanding of modularity and openness in the acquisition process. This work adds to the body of knowledge of MOSA, modularity, and open systems. As the DoD continues to adopt MOSA, one of the biggest challenges this praxis has led to identify is that while there is plenty of empirical evidence to suggest MOSA's success, it is currently highly difficult to actually measure MOSA and its effects. What to measure, how to measure, and with what metrics to measure remain the key challenge. Overall, there is a strong need for an assessment methodology and tools to better gauge the understanding and impacts modularity has on

its programs.

Some additional contributions this praxis is with respect to the bodies of knowledge in the fields of Engineering Management, Project Management and Systems Engineering disciplines. Since the DoD heavily relies on the use of robust Systems Engineering and Engineering methods, these practices are considered key contributors for successful system acquisition. Furthermore, the acquisition process is generally managed by project managers and the praxis will be of high interest to program managers managing open and modular system integration into their programs.

Hopefully, the data this praxis analyzed and collected will serve as a background to construct a more robust framework involving defense acquisition processes which will help reduce the cost and schedule overrun of all large human-engineered systems.

5.6 Recommendations for Future Research

Future research may include limiting the sample of studied programs to only certain stages and size of acquisition programs. Furthermore, as new acquisition programs use MOSA with a goal of increasing innovation and implementing COTS, there will be more data pertaining to the performance of MOSA and that will help in furthering the study of MOSA and its impact on major system lifecycles. Better measures and stronger modular assessment tools are needed to empower acquisition teams and leaders to better design and make decisions on programs ensuring that programs better meet cost and schedule targets.

Subsequent work could also focus on a comparison between whether the technical aspect or the business aspect of the MOSA strategy is a more effective approach. It could also be a combination of both the approaches that make it an attractive acquisition

strategy. Similarly, an important system engineering question is in which part of the lifecycle process should MOSA be introduced? Another interesting aspect is studying the Defense Acquisition Life Cycle Compliance Chart - DoDI 5000.02 which lays out all the decision points and milestones and understand where MOSA will have the greatest impact and understand if the chart should be modified to ensure programs are being measured to modular and open system capabilities. Research is also needed to determine how modular designs affect the verification and validation steps.

Future work could include detailed MOSA, cost, and schedule metrics on legacy projects to compare and analyze benefits of modularity and openness. “There is a clear need for support regarding the governance, culture, and strategic issues relating to the engineering design of modular systems as an operational mode” (Pakkanen, Juuti & Lehtonen 2018). One vital area of research is security and safety with respect to open and modular designs in defense systems. The shift from controlled platforms to open systems while providing greater performance and cost improvements, also has posed a significant data and technology challenge. With the advancement of remote and global teams and the boost to open system architectures, managing the balance between security, complexity, ethical, and legal challenges in defense systems is critical to understand and set guidelines to ensure there is no misuse. Modularity is not a one-time approach but a constant updating of standards and interfaces to ensure that the DoD’s acquisition is efficient and ahead of the technology curve on a global scale. Lastly, there is a need to quantify modularity in terms of hardware, software, and platforms and associate all this information to real value so as to understand how much modularity is helping and saving in the long run. With better metrics and stronger modular assessment tools, acquisition

teams and leaders will be able to better design and make decisions on programs ensuring that programs perform better using what is likely a tailored MOSA strategy.

5.7 Conclusion

The development and acquisition of defense systems is a complex process. As discussed in the praxis, there are many examples from commercial, academic, and governmental organizations that demonstrate modularity and open systems have proven to be successful. MOSA was introduced with the strategy to help combat ineffective acquisition processes and support those shortcomings by working towards a strategy that encourages systems to be able to evolve with mission needs and that can be more affordable than replacing older systems that are running well over their intended life cycles. Enhancing competition, expediting technology refresh, enabling cost savings from a design and development aspect, and improving standardization by facilitating data sharing, are all the benefits MOSA was intended to target. Through the efficient incorporation of modular and open standards, open systems will accelerate the acquisition life cycle while accommodating for rapidly changing demands and leveraging concurrent innovation to address cost and schedule factors. From the GAO reports, it is suspected that the main leading factor for programs that are not meeting cost and schedule targets is due to underestimation of the initial costs. The dearth of MOSA data at this time prohibited statistical testing, however qualitative analysis and early indications indicate that project performance is improving with the use of the MOSA strategy over the historical acquisition approach. DoD acquisitions are prone to risk and uncertainty. This uncertainty leads to an escalation on cost and time of the acquisition process. It is vital to highlight that MOSA is not the end result but rather an approach that can enable

innovation and is one of the many procurement strategies that can help with the DoD acquisition problem.

References

- Adetunji, Oluwatomi, et al. "Managing System Obsolescence via Multicriteria Decision Making." *Systems Engineering*, vol. 21, no. 4, 30 Apr. 2018, pp. 307–321., doi:10.1002/sys.21436.
- Aerospace Vehicle Systems Institute. (2017, March). Exponential Growth of System Complexity. Retrieved from <https://savi.avsi.aero/about-savi/savi-motivation/exponential-system-complexity/>
- Alan S. Williams, *Reinventing Canadian Defence Procurement: A View from the Inside* (Montreal: McGill-Queen's University Press, 2006), 23; Department of National Defence. *Perspectives on the Capital Equipment Acquisition Process* (Ottawa: Department of National Defence, 2006), B1.
- Alexander, K., Burnside, B., & Hurst, K. J. (2019, October 21). Recent Updates to the DoD Approach and Strategy to Implementing Modularity and Openness in Weapon Systems. Retrieved November 24, 2020, from https://ndiastorage.blob.core.usgovcloudapi.net/ndia/2019/systems/Wed_22366_Hurst.pdf
- Allaoui, S., Bourgault, M., & Pellerin, R. (2019). Business transformation frameworks: Comparison and industrial adaptation. *Journal of Enterprise Transformation*, 1-28. doi:10.1080/19488289.2019.1571538
- Amadeo, Kimberly. Why Military Spending Is More Than You Think It Is. *The Balance*. April 22, 2019. <https://www.thebalance.com/u-s-military-budget-components-challenges-growth-3306320>.
- Arthur Ollett and John Coleman, "Break the stove-piped stranglehold on capability with

- an open system approach,” *Journal of Battlefield Technology* 18, no. 3 (December 2015)
- Auger, M. (2016, February 04). *The Evolution of Defence Procurement in Canada* Background Paper. Retrieved December 06, 2020, from https://lop.parl.ca/sites/PublicWebsite/default/en_CA/ResearchPublications/201609E
- Azizian, Nazanin, et al. “A Framework for Evaluating Technology Readiness, System Quality, and Program Performance of U.S. DoD Acquisitions.” *Systems Engineering*, vol. 14, no. 4, 27 Apr. 2011, pp. 410–426., doi:10.1002/sys.20186.
- Badiru, A. B., & Thomas, M. U. (2013). Quantification of the PICK Chart for Process Improvement Decisions. *Journal of Enterprise Transformation*, 3(1), 1-15. doi:10.1080/19488289.2013.784221
- Baldwin, C. Y., & Clark, K. B. (2000). *Design rules (The Power of Modularity, v. 1)*. Cambridge: MIT Press.
- Baldwin, C. Y., & Clark, K. B. (2014, August 01). *Managing in an Age of Modularity*. Retrieved from <https://hbr.org/1997/09/managing-in-an-age-of-modularity>
- Bannasch, F., Grüntges, V., Rossi, G., & Weig, F. (2018, November 28). *Platforming and modularity: Smart answers to ever-increasing complexity*. Retrieved September 11, 2020, from <https://www.mckinsey.com/business-functions/operations/our-insights/platforming-and-modularity-smart-answers-to-ever-increasing-complexity>
- Barrett, Donald A, Luke A Borntrager, David M Green, “Blue Guardian: on open architecture for rapid ISR demonstration,” *Proceedings of SPIE 9849, Open*

- Architecture/Open Business Model Net-Centric Systems and Defense Transformation 2016, May 2016.
- Beckhusen, R. (2017, November 6). The B-58 Hustler Was a Beautiful Mistake. Retrieved from <https://warisboring.com/the-b-58-hustler-was-a-beautiful-mistake/>
- Bratton, J. (2019, November 20). Modular Open Systems Approach for weapons systems is a warfighting imperative. Retrieved December 5, 2020, from <https://militaryembedded.com/radar-ew/signal-processing/modular-open-systems-approach-for-weapons-systems-is-a-warfighting-imperative>
- Buiga, A. 2012. Investigating the Role of MQB Platform in Volkswagen Group's Strategy and Automobile Industry. *International Journal of Academic Research in Business and Social Sciences* 2 (9): 391–399.
- Chao, R. O., & Lenox, M. J. (2018). Towards a unified model of innovation and technological change. *Journal of Enterprise Transformation*, 1-16.
- Christianson, M. J. (2018, May 8). A Pragmatic Approach To Procurement: The Use of Modular Open Systems. Retrieved October 4, 2020, from <https://www.cfc.forces.gc.ca/259/290/405/305/christianson.pdf>
- Clark, D. (2019, September 13). The Future of Open Systems in Defence Procurement. Retrieved December 06, 2020, from <https://www.atkinsglobal.com/~media/Files/A/Atkins-Corporate/uk-and-europe/sectors-documents/defence/the-future-of-open-systems-in-defence-procurement.pdf>
- Craig J. Stone, "Improving the Acquisition Process in Canada," University of Calgary: School of Public Policy Research Papers 8, no. 16 (April 2015): 1; Ross Fetterly,

- Implementing Strong, Secure, Engaged: The Challenges Ahead (Calgary: Canadian Global Affairs Institute, January 2018)
- Daly, John J. "ACQUISITION SYSTEM INTEROPERABILITY CONSIDERATIONS." Systems Engineering Conference, National Defense Industrial Association (NDIA), 25 Oct. 2017, ndiastorage.blob.core.usgovcloudapi.net/ndia/2017/systems/Wednesday/Track2/19815_Daly.pdf.
- Danese, P., & Filippini, R. (2013). Direct and Mediated Effects of Product Modularity on Development Time and Product Performance. *IEEE Transactions on Engineering Management*, 60(2), 260-271. doi:10.1109/tem.2012.2208268
- Dasch, Jean M., and David J. Gorsich. "Survey of modular military vehicles: benefits and burdens." *Defense A R Journal*, vol. 23, no. 1, 2016, p. 2+. Accessed 11 Sept. 2020.
- Davis, S. A. (2011, November 2). Consolidated Afloat Networks and Enterprise Services (CANES) Fact Sheet. Retrieved November 24, 2020, from <https://www.secnv.navy.mil/rda/Documents/canes+overview+for+asn+rda+11-2-11-s.pdf>
- Department of Defense Directive 5000.01, The Defense Acquisition System (Nov. 2007); Department of Defense Instruction 5000.02, Operation of the Defense Acquisition System (Jan. 2015) (incorporating change 3 (Aug. 2017)) (DOD Instruction 5000.02).
- Devereaux, J. E. (2010). Obsolescence: A systems engineering and management approach for complex systems (Abstract). Massachusetts Institute of Technology.

- Retrieved July 16, 2019, from <https://dspace.mit.edu/handle/1721.1/59233>.
- DoD Best Practices for Using SE Standards on Contracts for DoD Acquisition Programs
April 2017; <http://www.acq.osd.mil/se/pg/guidance.html>
- Douglas A. Gregory, Parag Batavia, Mark Coats, Chris Allport, Ann Jennings, Richard Ernst, "The UAS control segment architecture: an overview," Proc. SPIE 8754, Open Architecture/Open Business Model Net-Centric Systems and Defense Transformation 2013, 875404 (29 May 2013); <https://doi-org.proxygw.wrlc.org/10.1117/12.2019836>
- Downing, C. (2017, June 15). The FACE™ Standard is now Ready and Open for Business. Retrieved September 27, 2020, from https://blogs.windriver.com/wind_river_blog/2017/06/the-face-standard-is-now-ready-and-open-for-business/
- Downing, C. (2018, May 30). Wind River and Rockwell Collins Bring Open Architectures from Planning to Production. Retrieved August 31, 2020, from https://blogs.windriver.com/wind_river_blog/2018/05/wind-river-and-rockwell-collins-bring-open-architectures-from-planning-to-production/
- Engel, A., & Browning, T. R. (2006, May). Designing systems for adaptability by means of architecture options. Retrieved from <http://onlinelibrary.wiley.com/doi/10.1002/sys.20090/full>
- Ernst, Rich, "Unmanned Systems (UxS) Control Segment (UCS) Architecture Overview," March 2016, <http://www.dtic.mil/ndia/2016/GRCCE/Ernst.pdf>.
- Falcone, S. J. (2018, February 6). Modular Open Systems Architecture/Approaches - Air Force LCMC. Retrieved December 05, 2019, from

- <http://www.ncmaboston.org/wp-content/uploads/2018/03/11.-Modular-and-Open-Architectures.ppt>
- Falcone, S. J. (2018, February 6). Modular Open Systems Architecture/Approaches - Air Force LCMC. Retrieved December 05, 2019, from <http://www.ncmaboston.org/wp-content/uploads/2018/03/11.-Modular-and-Open-Architectures.ppt>
- Fazlollahtabar, H., & Aliahmadi, A. (2014). Bayesian Dynamic Program for a New Product Development. *Journal of Enterprise Transformation*, 4(4), 329-344. doi:10.1080/19488289.2014.930542
- Freedberg, S. J. (2020, March 25). MOSA: The Invisible, Digital Backbone Of FVL. Retrieved December 17, 2019, from <https://breakingdefense.com/2020/03/mosa-fvls-invisible-digital-backbone/>
- Galvin, P. and Morkel, A. (2001), “The effect of product modularity on industry structure: the case of the world bicycle industry”, *Industry and Innovation*, Vol. 8 No. 1, pp. 31-47
- GAO Analysis of DoD provided data, DoD Instruction 5000.02, and best practices. GAO-18-360SP (Department of Defense Directive 5000.01)
- GAO, U. (2005, October 28). Program Evaluation: OMB's PART Reviews Increased Agencies' Attention to Improving Evidence of Program Results. Retrieved September 26, 2020, from <https://www.gao.gov/products/GAO-06-67>
- Gavin, V. (2009). Program Executive Office Littoral and Mine Warfare: Presentation to National Symposium on Sensor Fusion and Data Fusion (PowerPoint). Retrieved from <https://acc.dau.mil/adl/en->

- US/299408/file/45193/MSSA%20Data%20Fusion_final%207-31.pdf
- Gonzalez-Zugasti, J. P., K. N. Otto, and J. D. Baker. 1999. Assessing Value for Product Family Design and Selection. In Proceedings of the 25th Design Automation Conference, Las Vegas, NV, September 12–15.
- Government Accountability Office, March 2012, Report GAO 12-400SP, Defense Acquisitions Assessment of Selected Weapon Programs, Washington DC; pg 8.
- Government Accountability Office, U. (2013, July 31). Defense Acquisitions: DOD Efforts to Adopt Open Systems for Its Unmanned Aircraft Systems Have Progressed Slowly. Retrieved December 07, 2020, from <https://www.gao.gov/products/GAO-13-651>
- Government Accountability Office, U. (2018, April 25). WEAPON SYSTEMS ANNUAL ASSESSMENT. Retrieved from <https://www.gao.gov/assets/700/691473.pdf>
- Government Accountability Office. (n.d.). U.S. GAO - About GAO - Overview. Retrieved from <https://www.gao.gov/about/>
- Gu, P., and S. Sosale. 1999. Product Modularization for Life Cycle Engineering. Robotics and Computer-Integrated Manufacturing 15 (5): 387–401. doi: 10.1016/S0736-5845(99)00049-6
- Gu, P., M. Hashemian, and A. Y. C. Nee. 2004. Adaptable Design. CIRP Annals-Manufacturing Technology 53 (2): 539–557. doi: 10.1016/S0007-8506(07)60028-6
- Guertin, N. (2018, October 15). Emerging Opportunities in Modularity and Open Systems Architectures - First in a Series. Retrieved December 07, 2020, from

- https://insights.sei.cmu.edu/sei_blog/2018/10/emerging-opportunities-in-modularity-and-open-systems-architectures.html
- Henshaw, M. (2019, August 12). Assessment of open architectures within defence procurement issue 1: Systems of systems approach community forum working group 1 - open systems and architectures. Retrieved August 31, 2020, from https://repository.lboro.ac.uk/articles/Assessment_of_open_architectures_within_defence_procurement_issue_1_systems_of_systems_approach_community_forum_working_group_1_-_open_systems_and_architectures_/9559823
- Heydari, B., & Dalili, K. (2015). Emergence of Modularity in System of Systems: Complex Networks in Heterogeneous Environments. *IEEE Systems Journal*, 9(1), 223-231. doi:10.1109/jsyst.2013.2281694
- Heydari, Babak, et al. "From Modular to Distributed Open Architectures: A Unified Decision Framework." *Systems Engineering*, vol. 19, no. 3, 3 Aug. 2016, pp. 252–266., doi:10.1002/sys.21348.
- Hunt, G. (2011, August 18). Open for business: Generic vehicle architecture. Retrieved August 31, 2020, from <https://www.army-technology.com/features/feature127511/>
- INCOSE. 2015. *Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*, version 4.0. Hoboken, NJ, USA: John Wiley and Sons, Inc, ISBN: 978-1-118-99940-0
- Jacobs, M., Vickery, S. K., & Droge, C. (2007). The effects of product modularity on competitive performance. *International Journal of Operations & Production Management*, 27(10), 1046-1068. doi:10.1108/01443570710820620

- Jesús F. Lampón, Pablo Cabanelas & Javier González-Benito (2017) The impact of modular platforms on automobile manufacturing networks, *Production Planning & Control*, 28:4, 335-348, DOI: 10.1080/09537287.2017.1287442
- Keiser, E., & Edwards, J. (2017, August 08). Army aims for common, modular, open architecture. Retrieved December 06, 2020, from <https://www.c4isrnet.com/it-networks/2016/06/03/army-aims-for-common-modular-open-architecture/>
- Kendall, F. (2015). Implementation directive for Better Buying Power 3.0—Achieving dominant capabilities through technical excellence and innovation (Memorandum). Retrieved from [http://www.acq.osd.mil/fo/docs/betterBuyingPower3.0\(9Apr15\).pdf](http://www.acq.osd.mil/fo/docs/betterBuyingPower3.0(9Apr15).pdf)
- Kirkpatrick, David. (2004). Trends in the costs of weapon systems and the consequences. *Defence and Peace Economics*. 15. 259-273. 10.1080/1024269032000123203.
- Lopez, T. (2020, January 31). DOD to Require Cybersecurity Certification in Some Contract Bids. Retrieved December 06, 2020, from <https://www.defense.gov/Explore/News/Article/Article/2071434/dod-to-require-cybersecurity-certification-in-some-contract-bids/>
- Lorenzi, S. and Lello, A.D. (2001), “Product modularity theory and practice: the benefits and difficulties in implementation within a company”, *International Journal of Automotive Technology and Management*, Vol. 1 No. 4, pp. 425-48.
- Luo, J. (2017). System-of-systems innovation: Proactive methods for conception and strategies for implementation. *Journal of Enterprise Transformation*, 1-25. doi:10.1080/19488289.2017.1409296
- McCarthy, N. (2018, September 14). The Evolution Of The F-35's Unit Cost

- (Infographic). Retrieved from
<https://www.forbes.com/sites/niallmccarthy/2018/09/14/the-evolution-of-the-f-35s-unit-cost-infographic/#3635b04e7501>
- McCormick, R. (2018, January 22). Measuring the Impact of Sequestration and the Drawdown on the Defense Industrial Base. Retrieved December 07, 2020, from
<https://www.csis.org/analysis/measuring-impact-sequestration-and-drawdown-defense-industrial-base>
- Mesa, J. A., Esparragoza, I., & Maury, H. (2019). Modular architecture principles – MAPs: A key factor in the development of sustainable open architecture products. *International Journal of Sustainable Engineering*, 1-15.
doi:10.1080/19397038.2019.1634157
- Military.com. (2018, January). B-52 Stratofortress. Retrieved from
<https://www.military.com/equipment/b-52-stratofortress>
- Ministry of Defence, "National security through technology: technology equipment and support for UK defence and security", White Paper Cm8278, 2012.
- Minor, P. (2017, April 14). Identifying the Return on Investment for Army Migration to a Modular Open Systems Approach for Future and Legacy Systems. Retrieved November 24, 2020, from <https://apps.dtic.mil/sti/pdfs/AD1040338.pdf>
- Modular Open Systems Approach. DoD Systems Engineering - Initiatives, Office of the Deputy Assistant Secretary of Defense (ODASD), 7 Oct. 2017,
www.acq.osd.mil/se/initiatives/init_mosa.html.
- National Research Council, Physical Sciences, Computer Science and Telecommunications Board, & Committee for Advancing Software-Intensive

- Systems Producibility. (2010, October 22). Critical Code: Software Producibility for Defense. Retrieved from <https://www.nap.edu/catalog/12979/critical-code-software-producibility-for-defense>
- Navy, U. (2018, May). SPAWAR Technology. Retrieved from <https://www.public.navy.mil/spawar/technology/Pages/ConsolidatedAfloatNetworksandEnterpriseServicesCANES.aspx>
- O. Worthington, J. Connah, D. Chambers, A. Solomon and J. McInnes, "Openness in military systems part 2: Case studies," MILCOM 2012 - 2012 IEEE Military Communications Conference, Orlando, FL, 2012, pp. 1-6, doi: 10.1109/MILCOM.2012.6415652.
- Office of the Deputy Assistant Secretary of Defense. Systems engineering – Modular open systems approach, http://www.acq.osd.mil/se/initiatives/init_mosa.html (2017, accessed 30 June 2017).
- Pakkanen, J., Juuti, T., & Lehtonen, T. (2018). Identifying and addressing challenges in the engineering design of modular systems – case studies in the manufacturing industry. *Journal of Engineering Design*, 30(1), 32-61.
doi:10.1080/09544828.2018.1552779
- Pakkanen, J., Juuti, T., & Lehtonen, T. (2018). Identifying and addressing challenges in the engineering design of modular systems – case studies in the manufacturing industry. *Journal of Engineering Design*, 30(1), 32-61.
doi:10.1080/09544828.2018.1552779
- Parsons, Dan. "New Acquisition Structure Emphasizes Rapid Tech Insertion Through Open Architecture." *Defense Daily*, Defense Daily, 3 Aug. 2017,

- www.defensedaily.com/new-acquisition-structure-emphasizes-rapid-tech-insertion-open-architecture/pentagon/.
- Pellerin, C. (2014, November 5). Open Architecture Cuts Cost, Promotes Competition, Official Says. Retrieved December 6, 2020, from <https://www.defense.gov/Explore/News/Article/Article/603587/>
- Pocock, Chris. "U-2 Pioneers Open-mission Systems for US Air Force." AIN, 6 September 2015. <https://www.ainonline.com/aviation-news/defense/2015-09-06/u-2-pioneers-open-mission-systems-us-Air Force>
- Ramesh, N., & Delen, D. (2019). Why do enterprise transformations fail? HINT: IT'S not what you think. *Journal of Enterprise Transformation*, 1-16.
doi:10.1080/19488289.2019.1578838
- Reese, S. (2015, November 30). Developing Common Standards to Improve Interoperability. Retrieved December 06, 2020, from <http://www.armyaviationmagazine.com/index.php/archive/not-so-current/889-developing-common-standards-to-improve-interoperability>
- Ro, Y. K., Liker, J. K., & Fixson, S. K. (2007). Modularity as a Strategy for Supply Chain Coordination: The Case of U.S. Auto. *IEEE Transactions on Engineering Management*, 54(1), 172-189. doi:10.1109/tem.2006.889075
- S. Brabson and T. Anderson, "Evolution of the US Navy's collision avoidance systems (CAS) to future airborne capability environment (FACE)," in *IEEE Aerospace and Electronic Systems Magazine*, vol. 30, no. 6, pp. 16-23, June 2015, doi: 10.1109/MAES.2015.140075.
- Seawright, S. (2020, August 11). Modular Open Systems Architecture and its Impact on

- Military Weapon Systems Innovation. Retrieved December 06, 2020, from <https://modernbattlespace.com/2020/08/11/modular-open-systems-architecture-impact-military-weapon-systems-innovation/>
- Seawright, S. (2020, November 03). An Open Systems Approach in the Future Airborne Capability Environment in the Battlespace. Retrieved December 4, 2020, from <https://modernbattlespace.com/2020/11/03/open-systems-approach-future-airborne-capability-environment-battlespace/>
- Secretary of State for Defence, . (Ed.). (2012, February). National Security Through Technology: Technology, Equipment, and Support for UK Defence and Security. Retrieved August 31, 2020, from https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/27390/cm8278.pdf
- Serbu, J. (2019, February 11). In military services, new push from the top toward open systems architectures. Retrieved December 1, 2020, from <https://federalnewsnetwork.com/defense-main/2019/02/in-military-services-new-push-from-the-toward-open-systems-architectures/>
- Simonite, T. (2020, April 02). Moore's Law Is Dead. Now What? Retrieved August 31, 2020, from <https://www.technologyreview.com/2016/05/13/245938/moores-law-is-dead-now-what/>
- SPAWAR. (2011, July 27). CANES Program Successfully Achieves Critical Design Review. Retrieved from <https://www.doncio.navy.mil/ContentView.aspx?id=2532>
- STODDARD, R. (2018, April 16). Why Does Software Cost So Much? Retrieved from

- https://insights.sei.cmu.edu/sei_blog/2018/04/why-does-software-cost-so-much.html
- Strei, T. J. (2003, June 1). Open Architecture in Naval Combat System Computing of the 21st Century. Retrieved December 06, 2020, from <https://apps.dtic.mil/dtic/tr/fulltext/u2/a467321.pdf>
- Tokar, Joyce. (2017). An Examination of Open System Architectures for Avionics Systems – An Update,
- Ulrich, K. (1995). The role of product architecture in the manufacturing firm. *Research Policy*, 24(3), 419-440. doi:10.1016/0048-7333(94)00775-3
- Under Secretary of Defense, Implementation Directive for Better Buying Power 3.0 – Achieving Dominant Capabilities through Technical Excellence and Innovation, 9 Apr 2015, [http://www.acq.osd.mil/fo/docs/betterBuyingPower3.0\(9Apr15\).pdf](http://www.acq.osd.mil/fo/docs/betterBuyingPower3.0(9Apr15).pdf).
- Underwood, K. (2018, November 01). An Open-and-Shut Case for Army Open Architecture. Retrieved December 06, 2020, from <https://www.afcea.org/content/open-and-shut-case-army-open-architecture>
- Uppal, R. (2020, September 19). US Military thrust on modular open system standards (MOSA) based embedded computers including SOSA & FACE for future sensors like radars and EW systems, Weapons to Vehicles. Retrieved December 06, 2020, from <https://idstch.com/technology/electronics/us-military-adopting-embedded-computers-based-on-modular-open-system-standards-sosa-for-future-weapons-to-vehicles-and-electronic-warfare-systems/>
- Upton, D. M. (1995, August 01). What Really Makes Factories Flexible? Retrieved from <https://hbr.org/1995/07/what-really-makes-factories-flexible>

- Valerdi, Ricardo, et al. "Reliability Improvement of Major Defense Acquisition Program Cost Estimates-Mapping DoDAF to COSYSMO." *Systems Engineering*, vol. 18, no. 5, 17 Dec. 2015, pp. 530–547., doi:10.1002/sys.21327
- Vergun, D. (2020, August 20). DOD Developing Small, Unmanned Aerial System for Warfighters. Retrieved December 5, 2020, from <https://www.defense.gov/Explore/News/Article/Article/2318919/dod-developing-small-unmanned-aerial-system-for-warfighters/>
- Whittle, R. (2015, October 26). Lt. Gen. Otto Commits Air Force To More Open Mission Systems. Retrieved December 5, 2020, from <https://breakingdefense.com/2015/10/lt-gen-otto-commits-air-force-to-more-open-mission-systems/>
- Wilson, B. (2017, May 12). Improving Value of Strategic Defense Systems using Modular Open Architecture. Retrieved December 06, 2020, from <https://dspace.mit.edu/bitstream/handle/1721.1/111233/1003284216-MIT.pdf?sequence=1&isAllowed=y>
- Zimmerman, P., Ofori, M., Barrett, D., Soler, J., & Harriman, A. (2018). Considerations and examples of a modular open systems approach in defense systems. *The Journal of Defense Modeling and Simulation*. <https://doi.org/10.1177/1548512917751281>

Appendix A – Cost Analysis of Major Defense Acquisition Programs of 2018

The table below contains the data points that were used to evaluate Hypothesis 1. The table contains the cost analysis of all the 86 listed major defense acquisition programs of 2018.

Table A1: Cost Analysis of Major Defense Acquisition Programs of 2018 (GAO-18-360SP, 2018)

Sl. #	Program Name	First Estimate of Total Cost	December 2016 Total Cost	Change in USD Million	Percentage Change
1	Advanced Arresting Gear (AAG)	2,070.50	2,090.64	20.14	1.0%
2	AH-64E Apache New Build (AH-64E New Build)	2,621.89	1,942.38	-679.51	25.9%
3	AH-64E Apache Remanufacture (AH-64E Remanufacture)	8,012.96	14,093.65	6,080.69	75.9%
4	Air and Missile Defense Radar (AMDR)	6,184.71	5,707.89	-476.82	7.7%
5	Armored Multi-Purpose Vehicle (AMPV)	11,270.28	11,352.88	82.60	0.7%
6	AIM-120 Advanced Medium Range Air-to-Air Missile (AMRAAM)	12,094.04	25,176.44	13,082.40	108.2%
7	Airborne Warning and Control System Block 40/45 Upgrade (AWACS Blk 40/45 Upgrade)	3,088.98	2,968.37	-120.61	3.9%
8	Amphibious Combat Vehicle (ACV)	1,940.83	1,838.95	-101.88	5.2%
9	B-2 Defensive Management System Modernization (B-2	2,654.14	2,634.72	-19.42	0.7%

	DMS-M)				
10	B61 Mod 12 Life Extension Program Tailkit Assembly (B61 Mod 12 LEP TKA)	1,446.43	1,247.24	-199.19	13.8%
11	C-130J Hercules Transport Aircraft (C-130J)	1,050.01	17,496.05	16,446.04	1566.3%
12	C-5 Reliability Enhancement and Re-engining Program (C-5 RERP)	12,065.28	7,658.26	-4,407.02	36.5%
13	CH-47F Improved Cargo Helicopter (CH-47F)	3,562.39	16,191.50	12,629.11	354.5%
14	CH-53K Heavy Lift Replacement Helicopter (CH-53K)	18,318.54	28,318.75	10,000.21	54.6%
15	Columbia Class Ballistic Missile Submarine (SSBN 826)	102,225.52	101,277.54	-947.98	0.9%
16	Combat Rescue Helicopter (CRH)	8,595.45	8,746.84	151.39	1.8%
17	Common Infrared Countermeasure (CIRCM)	2,666.14	2,690.94	24.80	0.9%
18	Cooperative Engagement Capability (CEC)	3,256.49	6,455.94	3,199.45	98.2%
19	DDG 1000 Zumwalt Class Destroyer (DDG 1000)	38,505.53	24,492.81	-14,012.72	36.4%
20	DDG 51 Arleigh Burke Class Guided Missile Destroyer (DDG 51)	16,803.68	128,236.10	111,432.42	663.1%
21	E-2D Advanced Hawkeye Aircraft (E-2D AHE)	16,321.76	22,466.61	6,144.85	37.6%
22	EA-18G Growler Aircraft (EA-18G)	9,930.79	16,779.94	6,849.15	69.0%

23	Evolved Expendable Launch Vehicle (EELV)	19,224.09	57,889.90	38,665.81	201.1%
24	Enhanced Polar System (EPS)	1,475.73	1,462.66	-13.07	0.9%
25	F-15 Eagle Passive Active Warning Survivability System (F-15 EPAWSS)	4,414.39	2,690.30	-1,724.09	39.1%
26	F-22 Increment 3.2B Modernization (F-22 Inc 3.2B Mod)	1,657.97	1,530.58	-127.39	7.7%
27	F-35 Lightning II Program (F-35)	236,448.60	355,281.38	118,832.78	50.3%
28	Family of Advanced Beyond Line-of-Sight Terminals (FAB-T)	3,527.77	4,660.68	1,132.91	32.1%
29	MQ-8 (Fire Scout)	2,893.62	2,961.34	67.72	2.3%
30	Ground/Air Task Oriented Radar (G/ATOR)	1,621.80	2,916.27	1,294.47	79.8%
31	Global Broadcast Service (GBS)	637.21	1,366.74	729.53	114.5%
32	Guided Multiple Launch Rocket System/ Guided Multiple Launch Rocket Sys Alt Warhead (GMLRS/GMLRS AW)	1,956.41	7,734.43	5,778.02	295.3%
33	Global Positioning System III (GPS III)	4,360.81	5,772.47	1,411.66	32.4%
34	Next Generation Operational Control System (OCX)	3,663.35	5,569.33	1,905.98	52.0%
35	H-1 Upgrades (4BW/4BN) (H-1 Upgrades)	4,012.79	13,160.23	9,147.44	228.0%
36	HC/MC-130 Recapitalization Aircraft (HC/MC-130 Recap)	9254.32	14419.29	5,164.97	55.8%
37	Handheld, Manpack, and Small Form Fit	11,105.50	9,640.89	-1,464.61	13.2%

	Radios (HMS)				
38	Infrared Search and Track (IRST)	2,226.28	2,213.84	-12.44	0.6%
39	Integrated Air and Missile Defense (IAMD)	5,563.75	7,144	1,580.25	28.4%
40	Joint Air-to-Ground Missile (JAGM)	5,958.28	5,951.13	-7.15	0.1%
41	Joint Air-to-Surface Standoff Missile Extended Range (JASSM-ER)	2,494.60	4,361.40	1,866.80	74.8%
42	Joint Direct Attack Munition (JDAM)	3,782.65	11,330.94	7,548.29	199.6%
43	Joint Light Tactical Vehicle (JLTV)	24,931.82	21,369.69	-3,562.13	14.3%
44	Joint Precision Approach and Landing System (JPALS)	1,120.00	1,896.58	776.58	69.3%
45	KC-130J Transport Aircraft (KC-130J)	10,494.26	9,826.34	-667.92	6.4%
46	Littoral Combat Ship (LCS)	2,482.76	20,814.18	18,331.42	738.3%
47	Littoral Combat Ship - Mission Modules (LCS Packages)	7,250.26	7,291.28	41.02	0.6%
48	LHA 6 America Class Amphibious Assault Ship (LHA 6)	3,518.46	10,362.44	6,843.98	194.5%
49	LPD 17 San Antonio Class Amphibious Transport Dock (LPD 17)	12,958.90	23,371.75	10,412.85	80.4%
50	M109A7 Family of Vehicles (M109A7 FOV)	7,399.44	7,494.61	95.17	1.3%
51	M88A2 Heavy Equipment Recovery Combat Utility Lift Evacuation System (M88A2 Hercules)	3,330.18	3,280.91	-49.27	1.5%

52	MH-60R Multi-Mission Helicopter (MH-60R)	6,125.08	14,845.07	8,719.99	142.4%
53	MQ-1C Gray Eagle Unmanned Aircraft System (MQ-1C Gray Eagle)	1,123.25	6,064.89	4,941.64	439.9%
54	MQ-4C Triton Unmanned Aircraft System (MQ-4C Triton)	14,214.53	15,509.92	1,295.39	9.1%
55	MQ-9 Reaper Unmanned Aircraft System (MQ-9 Reaper)	2,917.49	13,042.91	10,125.42	347.1%
56	Multifunctional Information Distribution System (MIDS)	1,442.90	5,205.16	3,762.26	260.7%
57	Navy Multiband Terminal (NMT)	2,567.96	2,439.52	-128.44	5.0%
58	Next Generation Jammer Increment 1 (NGJ Inc 1)	7,749.66	7,762.64	12.98	0.2%
59	P-8A Poseidon Multi-Mission Maritime Aircraft (P-8A)	34,336.44	35,044.89	708.45	2.1%
60	Standard Missile-6 (SM-6)	6,306.77	9,674.83	3,368.06	53.4%
61	SSN 774 Virginia Class Submarine (SSN 774)	66,881.28	137,099.81	70,218.53	105.0%
62	Tactical Tomahawk RGM-109E/UGM 109E Missile (TACTOM)	2,341.40	7,518.01	5,176.61	221.1%
63	Trident II (D-5) Sea-Launched Ballistic Missile UGM 133A (Trident II Missile)	52,233.71	59,636.42	7,402.71	14.2%
64	UH-60M Black Hawk Helicopter (UH-60M Black Hawk)	14,351.28	28,012.86	13,661.58	95.2%

65	V-22 Osprey Joint Services Advanced Vertical Lift Aircraft (V-22)	44,371.15	65,076.53	20,705.38	46.7%
66	Warfighter Information Network-Tactical Increment 3 (WIN-T Inc 3)	18,109.63	2,077.79	-16,031.84	88.5%
67	Advanced Extremely High Frequency (AEHF) Satellite	7,048.33	15,276.77	8,228.44	116.7%
68	AGM-88E Advanced Anti-Radiation Guided Missile (AGM-88E AARGM)	1,771.05	2,784.14	1,013.09	57.2%
69	AIM-9X Block II Sidewinder (AIM-9X Blk II)	4,420.39	3,452.03	-968.36	21.9%
70	Airborne and Maritime/Fixed Station (AMF)	9,021.63	3,321.05	-5,700.58	63.2%
71	Chemical Demilitarization-Assembled Chemical Weapons Alternatives (Chem Demil-ACWA)	2,923.24	13,547.63	10,624.39	363.4%
72	Gerald R. Ford Class Nuclear Aircraft Carrier (CVN 78)	39,359.85	39,661.55	301.70	0.8%
73	Intercontinental Ballistic Missile Fuze Modernization (ICBM Fuze Mod)	1,927.97	1,926.48	-1.49	0.1%
74	Integrated Defensive Electronic Countermeasures (IDECM) Blocks 2/3/4	2,408.84	3,073.14	664.30	27.6%
75	KC-46 Tanker Modernization Program (KC-46A)	48,488.25	41,171.84	-7,316.41	15.1%
76	Military GPS User Equipment (MGUE)	1,535.80	1,178.80	-357.00	23.2%

	Increment 1				
77	Mobile User Objective System (MUOS)	7,436.50	6,982.89	-453.61	6.1%
78	Offensive Anti-Surface Warfare Increment 1 (OASuW Inc 1)	1,558.80	1,632.85	74.05	4.8%
79	Patriot Advanced Capability-3 Missile Segment Enhancement (PAC-3 MSE)	8,062.34	6,882.40	-1,179.94	14.6%
80	Space Based Infrared System High (SBIRS High)	3,930.24	3,464.31	-465.93	11.9%
81	Small Diameter Bomb Increment II (SDB II)	5,202.30	4,390.29	-812.01	15.6%
82	Space Fence Ground-Based Radar System Increment 1	1,664.72	1,570.17	-94.55	5.7%
83	Ship to Shore Connector Amphibious Craft (SSC)	4,373.93	4,622.51	248.58	5.7%
84	VH-92A Presidential Helicopter Replacement Program	4,939.13	4,859.34	-79.79	1.6%
85	Wideband Global SATCOM (WGS)	1,320.83	4,177.63	2,856.80	216.3%
86	Warfighter Information Network-Tactical Increment 2 (WIN-T Inc 2)	4,102.63	11,203.43	7,100.80	173.1%